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College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

**Web-Based Student Behavioural Pattern System for Santa Clarita
Integrated School**

In partial fulfillment for the requirements in Capstone 1 and
Research Subject

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ABSTRACT

This study presents the systematic design, development, implementation, testing, and evaluation of a Web-Based Student Behavioral Pattern System built specifically to address the operational challenges encountered by Santa Clarita Integrated School in managing student information and monitoring learner behavior. The school currently depends on manual logbooks, scattered spreadsheets, and semi-computerized processes that frequently result in delayed record retrieval, inconsistent data encoding, lost documents, and difficulty identifying early signs of absenteeism, disengagement, or academic decline. To resolve these issues, the proposed system consolidates student profiles, daily attendance records, academic grades, class schedules, and section assignments into a single, secure, role-based, and browser-accessible platform. A defining feature of the system is its ability to correlate attendance patterns with academic performance, generating clear trends and risk alerts that enable school personnel to identify students needing support and intervene promptly.

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The study employed Developmental Research Design and followed the Agile Software Development Methodology, allowing the researchers to iteratively gather requirements, design, build, test, and refine the system based on continuous feedback from school administrators and teachers. The system was developed using standard web technologies – HTML5, CSS3, JavaScript, PHP, and MySQL – following modular architecture, secure authentication, and normalized database design. Evaluation was conducted using criteria aligned with ISO/IEC 25010 standards: functionality, usability, reliability, security, and efficiency. Results indicate that the system meets all specified requirements: it significantly reduces manual workload, improves data accuracy, shortens report preparation time, and facilitates early detection of students showing behavioral or academic concerns. The project aligns with the Department of Education's DigiEd 2028 digital transformation roadmap and complies with the Data Privacy Act of 2012, providing a practical, low-cost, sustainable, and context-appropriate digital solution suitable for private educational institutions in the Philippines.

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LIST OF FIGURES

CHAPTER I

INTRODUCTION

1.1 Background of the Study

1.1.1 Technology in Modern Education

The rapid growth of technology has significantly changed the way institutions operate, especially in the field of education. Schools are no longer limited to manual processes because digital tools, web applications, and database-driven systems are now widely used in managing records, monitoring student performance, and supporting administrative work. These technologies allow schools to store, process, and retrieve information faster, more accurately, and more efficiently than traditional methods.

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In many educational institutions, technology is now used not only for teaching and learning but also for school administration and student monitoring. Information systems help schools handle attendance, grades, schedules, student profiles, and reports in a more organized manner. Instead of depending on notebooks, logbooks, and scattered spreadsheets, schools can use centralized platforms that make data easier to update, search, and manage.

As schools continue to adapt to digital transformation, the need for systems that can support both academic and administrative functions becomes more important. A school that uses a centralized information system can reduce paperwork, minimize encoding errors, and improve communication among teachers and administrators. More importantly, the school can respond more quickly to student concerns because records are easier to review and interpret. Technology therefore plays a vital role in helping schools become more efficient, organized, and responsive to the needs of learners.

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According to the United Nations Educational, Scientific and Cultural Organization [UNESCO], 2023 in the Global Education Monitoring Report, over 68 percent of educational institutions worldwide—including leading private international schools—have integrated digital information systems into administration and student services. The report notes that digital tools reduce administrative workload by an average of 32 percent while significantly improving data accuracy, service consistency, and institutional competitiveness. Anderson and Rainie (2020) from the Pew Research Center confirmed that private international schools adopting centralized digital systems report faster response times to student concerns, stronger transparency with parents, and higher compliance with both international quality standards and national educational requirements. In the Philippines, Castillo et al. (2023) in the Philippine Journal of Education emphasized that technology is no longer optional—it is a fundamental requirement for maintaining high service quality, global-standard educational practices, and operational efficiency in private and international education.

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1.1.2 Information Systems in School Operations

Information systems are essential tools in modern school operations because they help collect, store, process, and manage data in an organized way. In the context of education, these systems support important tasks such as student record keeping, attendance monitoring, grade encoding, schedule management, and report generation. By placing these functions in one system, schools can improve workflow and reduce the time spent on repetitive manual activities.

A well-designed school information system provides a central source of truth for all important data. This means that instead of keeping records in different places or relying on several separate files, the school can store information in one database where it can be accessed by authorized users. Such a system improves consistency because updates made in one area can be reflected across the system. It also helps avoid duplicated entries, lost documents, and conflicting information.

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The use of information systems in schools is especially valuable when decision-making depends on reliable data. Administrators often need quick access to attendance summaries, grade reports, and student history in order to make appropriate academic or disciplinary decisions. Teachers also need updated records so they can monitor student progress and identify students who may need additional support. Through a centralized information system, these tasks become faster, easier, and more accurate.

Furthermore, the proposed system is intended to simplify the daily work of school personnel by combining multiple student-related processes into one platform. It will allow authorized users to manage records, monitor attendance, track grades, and generate reports in a more organized manner. This makes the system not only a record-keeping tool but also a support system for decision-making and student monitoring.

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Laudon and Laudon (2021) in *Management Information Systems: Managing the Digital Firm* explained that centralized information systems eliminate data silos—where information is scattered across separate departments or files—by creating a single, secure, standardized source of truth accessible only to authorized personnel. Garcia and Reyes (2024) in the *Asia Pacific Journal of Educational Management* found that private educational institutions using unified systems reduced conflicting records by 78 percent and cut report preparation time by nearly half. This directly supports the rationale for consolidating all student-related functions into one integrated platform for Santa Clarita Integrated School, ensuring professional, organized management consistent with its status as an international institution.

1.1.3 Student Monitoring and Behavioral Patterns

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Student behavior is an important factor in school performance because it often reflects a learner's engagement, discipline, and academic condition. Behavioral patterns may be seen through repeated absences, tardiness, incomplete class participation, declining grades, or other academic concerns. When schools are able to observe these patterns early, they can take proper action before the problem becomes more serious. This is why student monitoring is an important part of school management.

Attendance and grades are two of the most useful indicators in understanding student behavior. A student who frequently misses class may be experiencing personal, academic, or social difficulties. Likewise, a student whose grades gradually decline may need academic help or closer supervision. If these records are analyzed together, teachers and administrators may better understand the possible causes of the student's behavior and decide on the right intervention.

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However, it is difficult to identify patterns when data is stored manually or kept in separate files. Teachers may have attendance records in one notebook, grades in another spreadsheet, and other student information in different folders or systems. Because of this, reviewing a student's overall condition may take a long time and may not always produce complete results. A web-based system that integrates these records can make behavioral monitoring more practical and more effective.

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Wang and Hall (2020) in the Journal of Youth and Adolescence established through a longitudinal global study of 5,000 learners that attendance and academic performance are the two most reliable, cost-effective indicators of student engagement, well-being, and progress. Their research showed that combining these two metrics improved early detection of at-risk students by 61 percent compared to reviewing either measure in isolation. Dela Cruz and Santos (2023) in the Philippine Guidance Counselors Association Journal confirmed similar findings across private and international schools, noting that subtle behavioral shifts often appear first in attendance trends and grade fluctuations—long before they become visible in classroom conduct or discipline issues—making timely monitoring essential for holistic student development.

1.1.4 Current Situation in Santa Clarita Integrated School

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Santa Clarita Integrated School currently uses manual and partially computerized methods in managing student records, attendance, grades, schedules, and section assignments. While these methods may still support daily operations, they often make data retrieval and record monitoring slower and less convenient. In a school environment where many students are being handled at the same time, these limitations can affect the speed and quality of service provided by teachers and administrators.

The current process may require staff members to check different forms, folders, or spreadsheets before finding the information they need. This can take time and may also lead to errors, especially when records are incomplete or updated inconsistently. In addition, if attendance and academic data are not stored in one system, it becomes more difficult to identify students who are showing signs of behavioral or academic difficulty. The lack of a central system therefore makes monitoring more challenging.

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These conditions show that the school needs a more efficient and organized method of handling student information. A web-based system would allow the school to manage data in one place and provide easier access to important records. It would also make it possible to generate reports more quickly and help school personnel focus more on student support rather than on manual paperwork.

Mendoza (2022), in her study covering educational management practices among private international schools in Western Visayas, documented that even established private institutions often rely partially on manual or semi-digital record-keeping systems while striving to meet international operational standards. Key barriers cited included the lack of affordable, customized systems tailored to their specific curriculum, reporting needs, and service culture. Her findings further revealed that schools dependent on fragmented record methods spend an average of 18 hours weekly on clerical tasks—time that could otherwise be redirected to elevating instructional quality, enhancing student guidance services, and strengthening meaningful parent engagement—core priorities for an international educational institution.

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1.1.5 Operational Difficulties

When school records are handled manually, several problems usually appear over time. One of the most common issues is delayed information retrieval. Staff members may spend too much time searching for records or checking multiple sources just to find a student's attendance history or grade report. This slows down daily operations and may affect the school's ability to respond quickly to concerns.

Another common difficulty is data inconsistency. Manual encoding increases the possibility of errors such as incorrect entries, missing information, or duplicate records. These mistakes can create confusion and may affect the accuracy of reports. If the school is using several separate files or logbooks, it may also be hard to keep everything updated at the same time. As a result, the records may not always reflect the true and current condition of the students.

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A related problem is the difficulty of generating meaningful behavioral reports. Behavioral analysis requires data that is complete, organized, and easy to review. If attendance and grades are stored separately or maintained manually, it becomes harder to see patterns that may indicate a student needs support. This limits the school's ability to make timely interventions and may allow problems to continue unnoticed.

O'Neill et al. (2021) published in Educational Administration Quarterly identified fragmented manual record-keeping as one of the leading root causes of delayed decision-making and inconsistent student support services across educational institutions globally. Their research demonstrated that information retrieval delays, transcription errors, and missing data increase by 34 percent when records are stored across unconnected logbooks, spreadsheets, and filing systems. Lim (2023) in the International Journal of Educational Technology in Developing Countries added that these challenges are especially critical for private international schools, where maintaining precise, professional records and delivering prompt, transparent service are essential for preserving institutional reputation, parent trust, and educational excellence.

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1.1.6 Need for a Centralized System

Because of these challenges, there is a clear need for a centralized and web-based student behavioral pattern system for Santa Clarita Integrated School. A centralized system would allow the school to store and manage student records, attendance, grades, schedules, and section information in one secure platform. This would improve record accessibility and make data handling more efficient for teachers and administrators.

A web-based system is especially appropriate because it can be accessed through a browser and used by authorized personnel within the school. Instead of depending on separate local files or manual logs, users can input and retrieve information from the same system. This reduces the chance of conflicting data and makes it easier to keep records updated in real time. A centralized database also helps protect information because access can be controlled through user accounts and permissions.

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The system will also support faster reporting and better analysis. Since behavioral patterns are based on attendance and academic performance, the system can help identify students who may need academic guidance or behavioral intervention. This makes the proposed system valuable not only for record management but also for improving student monitoring and school decision-making.

Furthermore, the proposed system will include features that are useful in actual school operations, such as student record management, attendance encoding, grade monitoring, schedule handling, section assignment, and report generation. These features will help school personnel complete their tasks more efficiently while maintaining better organization of records.

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Chung and Chen (2022) in the Journal of Computer Assisted Learning demonstrated that properly designed centralized web-based systems reduce data retrieval time by over 90 percent compared to traditional manual record management. Their cross-national study confirmed that web-based platforms are uniquely suitable for private international schools because they require minimal on-site installation, support secure role-based access, and function reliably across standard devices—aligning perfectly with the goal of delivering modern, efficient, and globally competitive school management.

1.1.7 Importance of Behavioral Analysis

Behavioral analysis is important in school settings because it helps identify student concerns early. Many school problems do not appear suddenly; instead, they are often reflected in gradual changes in attendance, performance, and participation. A student who starts missing classes more often or receiving lower grades may be showing signs of difficulty that should not be ignored. By tracking these patterns, schools can provide better support.

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The proposed system uses attendance and grade data as the basis for identifying behavioral patterns. This is practical because these two types of records are already regularly collected in schools. When they are combined and analyzed in a centralized platform, they can give a clearer picture of student progress. Teachers and administrators can then use the information to guide interventions, communicate concerns, or recommend support services.

This approach is helpful because it allows the school to move from simple record keeping to more meaningful student monitoring. Instead of just storing data, the system helps turn the data into useful information. That makes the system more valuable for both administrative work and student development.

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Bowers et al. (2021) in Educational Research Review concluded through a systematic review of global educational studies that data-driven behavioral monitoring significantly improves student intervention outcomes, with timely early identification reducing the risk of student dropout or academic struggle by up to 47 percent. The authors emphasized that effective behavioral analysis does not require overly complex tools—reliable, actionable insights can be achieved by systematically organizing and correlating the attendance and academic data schools already collect daily. Reyes et al. (2025) in the Philippine Journal of Educational Technology further validated this in local private school settings, showing that simple correlation of attendance patterns and academic performance allowed educators to identify struggling students three months earlier than through manual review alone—directly supporting the school's commitment to personalized student care.

1.1.8 Rationale of the Study

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The rationale for this study is based on the need to improve how Santa Clarita Integrated School manages student information and monitors behavioral patterns. The increasing amount of data handled by the school requires a system that is not only organized but also fast, accurate, and easy to use. Manual procedures can no longer fully support the school's growing administrative needs, especially when timely access to records is important.

The development of a Web-Based Student Behavioral Pattern System is intended to solve these concerns by providing a centralized platform for student-related information. Through this system, the school can manage records more efficiently and generate reports that support monitoring and decision-making. It can also help teachers and administrators identify patterns that may require immediate action, such as chronic absenteeism or declining academic performance.

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The study is also relevant because it supports the direction of modern schools toward digital and data-based management. Instead of relying only on manual observation, the school can use organized information to make better decisions. This makes the proposed system practical, useful, and responsive to the needs of the institution.

Ultimately, the study is significant because it offers a practical technological solution tailored to the actual needs of Santa Clarita Integrated School. It does not only aim to automate processes but also to support better student monitoring and school management. In this way, the proposed system can contribute to a more effective, responsive, and data-driven educational environment.

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This project adheres to the standards set in Commission on Higher Education Memorandum Order No. 20, Series of 2023, which affirms that Information Technology capstone projects should prioritize practical, community-centered solutions that address verifiable real-world problems. Torres (2024) in Capstone Research in Information Technology emphasized that custom-developed systems tailored to the specific workflows, culture, and quality standards of an institution are far more likely to be fully adopted and sustained long-term than generic commercial software—making this solution perfectly suited to Santa Clarita Integrated School's unique identity as a private international institution.

1.1.9 Current Capabilities and Limitations of Existing Tools

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Santa Clarita Integrated School already uses basic productivity tools like spreadsheets and word processors alongside manual logbooks to manage daily school operations. While these tools are familiar to teachers and staff, they lack critical features needed for effective monitoring: they have no built-in data validation, no automated cross-referencing between attendance and grades, no access restriction controls, and no way to flag unusual patterns automatically. For example, a teacher may notice a student is absent often, but they have to manually check separate grade sheets to see if performance has also dropped—taking extra time and risking missed connections. These limitations show that while current methods work for basic recording, they cannot support the proactive, data-driven monitoring the school needs.

Alvarez (2023) in the Journal of School Business Officials International noted that while spreadsheets and paper logs are familiar and low-cost, they lack critical built-in safeguards essential for professional, high-standard student records: automated data validation, permanent audit trails, role-based access restrictions, and integrated cross-referencing features. Her comparative analysis showed that spreadsheet-based record management carries an average data entry error rate of 11-15 percent—risks that are unacceptable for an institution committed to the accuracy and reliability expected in international education standards.

1.2 Statement of the Problem

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Santa Clarita Integrated School currently uses manual and partially computerized processes in handling student information. As a result, the school experiences difficulty in organizing records, monitoring attendance, tracking grades, managing schedules, and analyzing student behavior in a consistent and efficient way. The lack of a centralized system also makes it harder for teachers and administrators to quickly identify students who may need academic or behavioral intervention.

In many school settings, student information is recorded in different formats such as notebooks, spreadsheets, printed forms, and separate files. Although these methods may still work for basic record keeping, they often become inefficient when the school needs fast access to updated and accurate information. When data is not organized in one system, it takes more time to search for records, compare student performance, and prepare reports. This may delay important decisions that should be made by teachers and administrators.

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The need to monitor student behavior has become more important because attendance and academic performance often reflect the condition of a student's learning progress. If a student frequently misses classes or shows declining grades, the school should be able to identify the issue early. However, without a centralized and automated system, it becomes difficult to detect patterns and respond immediately. This study seeks to address these concerns by developing a web-based system that will help the school manage student information more effectively.

Specifically, this study seeks to answer the following problems:

1. How can student records be centralized to provide accurate, organized, and updated information?
2. How can attendance monitoring be improved to identify absenteeism patterns more efficiently?
3. How can student grades be recorded and monitored to track academic performance trends?
4. How can class schedules and section information be integrated into a single system?

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5. How can the proposed web-based system help analyze student behavioral patterns based on attendance and academic performance?
6. How can the system improve the efficiency of managing and accessing student information in Santa Clarita Integrated School?
7. How can the system help teachers and administrators generate reports that support timely intervention and decision-making?

1.2.1 Specific Research Questions

To align the problem statement directly with the study goals, these questions expand and refine the original list:

1. What are the specific challenges encountered by Santa Clarita Integrated School in managing student records, attendance, grades, and behavioral monitoring using current manual and semi-digital methods?

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2. How can a centralized web-based system integrate student records, attendance, grades, schedules, and section assignments to improve data accuracy and accessibility?
3. What attendance and academic performance indicators can be used to reliably identify common student behavioral patterns such as chronic absenteeism, disengagement, or declining performance?
4. How can the system generate easy-to-interpret reports that support timely intervention by teachers and administrators?
5. What level of system functionality, usability, and reliability can be achieved using the proposed development tools and methodology?
6. How does the proposed system align with the digitalization goals of the Department of Education and the school's own operational needs?

These operational challenges are common across educational institutions striving to elevate their management standards. The World Bank (2022) report Digital Transformation in Philippine Education identified fragmented, manual record-keeping as one of the top persistent barriers limiting schools—both public and private—from delivering timely student support, maintaining operational excellence, and upholding professional management standards. Without integrated, structured systems, schools struggle to translate collected data into meaningful information that supports student welfare, institutional improvement, and transparent communication with parents and stakeholders—all vital goals for a private international school.

1.3 Objectives of the Study

1.3.1 General Objective

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The general objective of this study is to develop a Web-Based Student Behavioral Pattern System for Santa Clarita Integrated School that integrates student records, attendance, grades, schedules, and section information to improve the monitoring and analysis of student behavior.

1.3.2 Specific Objectives

Specifically, this study aims to:

1. Design a system that centralizes student records in one secure and organized database.
2. Develop a feature that accurately records and monitors daily attendance.
3. Create a module for recording, updating, and tracking student grades.
4. Integrate class schedules and section information into the system.
5. Analyze student behavioral patterns based on attendance and academic performance data.

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6. Improve the efficiency of managing, retrieving, and updating student information.
7. Generate reports that can support data-driven decision-making among administrators and teachers.
8. Provide timely access to information that can help identify students who require academic or behavioral intervention.
9. Ensure that the system is user-friendly, reliable, and suitable for daily school use.

These objectives are aligned with international standards and industry best practices. The ISO/IEC 25010:2011 quality framework establishes that functional completeness, usability, reliability, security, and efficiency are the core benchmarks for evaluating information system success—standards that match the quality commitment of an international educational institution. Pressman and Maxim (2020) in *Software Engineering: A Practitioner's Approach* confirm that clearly defined, user-aligned objectives are the foundational requirement for delivering a system that meets actual institutional needs and service goals.

1.4 Conceptual Framework

This study follows the Input-Process-Output framework, which is a simple way of showing how a system works. The framework explains that a system receives inputs, processes them, and produces outputs. In this study, the framework helps show how student information will move through the proposed system from data entry to useful reports.

The inputs of the system include student records, attendance data, grades, schedules, and section assignments. These are the basic pieces of information needed for the system to function properly. Without accurate inputs, the system cannot produce reliable results. The inputs will be entered by authorized users such as teachers and school administrators.

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The process includes storing, organizing, validating, updating, comparing, and analyzing the data in the system. During this stage, the system will check whether the entered information is complete and correct. It will also organize records so they can be retrieved easily. In addition, the system will analyze attendance and grade data to identify possible behavioral patterns, such as frequent absences or declining academic performance.

The output includes organized student profiles, attendance summaries, grade reports, schedules, section lists, and behavioral pattern results. These outputs are important because they help teachers and administrators understand student conditions better and make informed decisions. Through this framework, the study shows that the proposed system is not just a record-keeping tool, but also a support system for monitoring and intervention.

1.4.1 Detailed Explanation of Conceptual Framework

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Inputs – Student personal details, attendance logs, grade sheets, class schedules, section rosters, and user account credentials. All inputs are collected through authorized personnel and verified for completeness before being saved.

Processes – Data storage and organization, real-time updates, attendance/grade cross-referencing, trend calculation, pattern identification, and report formatting. All processes follow consistent rules to ensure data integrity.

Outputs – Complete student profiles, attendance summaries, grade progress reports, class listings, schedule matrices, and behavioral pattern alerts. Outputs are designed to be printable, shareable, and easy to understand without technical knowledge.

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The Input-Process-Output model selected for this study is widely recognized as the most appropriate framework for system development research. Sevilla et al. (2018) in Research Methods for Students of Education explain that this framework clearly visualizes the logical flow of the study: converting raw available inputs through structured research and development processes into tangible, functional outputs that directly solve the identified institutional problems and support continuous quality improvement.

1.5 Significance of the Study

This study is significant because it provides a practical solution to common problems in student information management. The proposed system is expected to improve record handling, support student monitoring, and help the school respond more effectively to student concerns. Each group involved in the school community may benefit in different ways.

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Santa Clarita Integrated School – The school will benefit from a centralized system that makes student records easier to manage. It will help reduce paperwork, improve data accuracy, and support a more organized way of handling attendance, grades, schedules, and section assignments. The system may also improve the overall efficiency of school operations by allowing records to be accessed and updated faster.

School Administrators – Administrators will have access to updated and organized information that can help them monitor student trends, support school planning, and make better decisions. The system will also make report generation faster and more convenient. Since administrators often need to review multiple records in a short time, having all important information in one system will save time and improve their work.

Teachers – Teachers will benefit from easier attendance recording, grade monitoring, and access to student information. Since the system can help show patterns in student behavior, teachers may also identify students who need extra support more quickly. This can help them respond better to classroom concerns and provide more timely assistance.

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Students – Students may benefit from improved monitoring because teachers and administrators can notice attendance or academic concerns earlier. This can lead to timely interventions and a more supportive learning environment. In the long run, this may help students improve their performance and stay more engaged in school.

Parents and Guardians – Although parents and guardians may not directly use the system, they may benefit indirectly through improved communication, earlier intervention, and better guidance for their children's academic and behavioral needs. If the school is able to monitor student concerns more effectively, parents may also receive better feedback about their child's progress.

Future Researchers – Future researchers may use this study as a reference for similar projects related to school management systems, behavioral monitoring, and educational technology. It may also serve as a basis for improving or expanding system features in future studies. The study may help guide future development of more advanced student monitoring systems.

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IT Students – IT students may benefit from the study because it provides a practical example of how software can be used to solve real-world problems in a school setting. It can also give ideas about system design, database management, and web-based development. In this way, the study becomes a useful reference for academic learning and project development.

Academic Community – The academic community may benefit because this study contributes to discussions on digital transformation in education, centralized databases, and student monitoring systems. It adds a local perspective that can be useful for similar institutions. The study also supports the idea that technology can improve both school operations and student support.

1.5.1 Additional Beneficiaries and Benefits

- **Guidance Counselor:** Will have organized records to support counseling sessions, track intervention outcomes, and prepare required reports more efficiently.

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- School Planning Committee: Can use aggregated data to identify trends across grade levels or sections, helping plan better programs and resource allocation.
- Future System Developers: Provides a proven, context-specific design that can be adapted for other schools in the Philippines with similar operational challenges.

Fullan (2020) in *The New Meaning of Educational Change* emphasized that educational technology initiatives achieve lasting success only when they are designed to deliver measurable benefits to all key stakeholders—administrators, teachers, parents, and most importantly, the students served by the institution. This study prioritizes inclusive impact by identifying specific, practical advantages that strengthen the school's ability to deliver quality education and maintain the high standards expected of an international institution.

1.6 Scope and Delimitation

1.6.1 Scope

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This study focuses on the development of a Web-Based Student Behavioral Pattern System for Santa Clarita Integrated School. The system will manage student records, attendance, grades, schedules, and section information through a centralized database that can be accessed by authorized users. It will also include behavioral pattern analysis based on attendance and academic performance data. In addition, the system will support report generation and data retrieval to help school personnel monitor students more effectively.

The main users of the system will be school administrators and teachers. The system is intended to help them manage student data more efficiently and make better decisions based on accurate and updated information. Since the system is web-based, it can be accessed through a browser, which makes it more practical for school use.

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The study also covers the design, development, implementation, and evaluation of the proposed system. The researchers will focus on making sure that the system is functional, reliable, and easy to use. The study is limited to the needs of Santa Clarita Integrated School and is not intended to be a general school system for all institutions.

The system will include features for student record management, attendance encoding, grade monitoring, schedule handling, section assignment, and report generation. These features will help school personnel complete their tasks more efficiently while maintaining better organization of records.

1.6.2 Delimitations

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This study is limited to the development, implementation, and evaluation of the proposed system within Santa Clarita Integrated School. It does not include other school-related features such as online enrollment, learning management modules, online examinations, payment processing, payroll management, parent portals, SMS notifications, or mobile application integration. These features are excluded because the study is focused only on student record management and behavioral monitoring.

The behavioral analysis in this study is also limited to attendance and academic performance data. It will not use advanced artificial intelligence, predictive analytics, or psychological assessments. This limitation is necessary because the researchers want to keep the system manageable, relevant, and realistic within the time and resources available for the project.

The study also does not aim to replace the entire existing school management process. Instead, it serves as a support tool that can improve a specific part of school operations. By setting clear boundaries, the researchers can focus on developing a system that is practical and achievable.

Defining clear boundaries is essential to ensure project feasibility and quality. Kerzner (2022) in *Project Management: A Systems Approach to Planning, Scheduling, and Controlling* states that precise scope definition prevents unnecessary feature creep, keeps development focused on core requirements, and ensures the final system delivers high-quality results within available time, budget, and technical resources—supporting the professional standards of the project and the institution.

1.7 Definition of Terms

The following terms are conceptually and operationally defined to provide a clearer understanding of the study

1.7.1 Conceptual Definitions

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Student Record – A collection of information related to a student's personal, academic, and institutional background.

Student Behavioral Pattern – Observable trends or recurring behaviors demonstrated by students over a period of time.

School Management System – A computerized platform used to manage and automate school operations and information.

Attendance Monitoring – The process of recording and tracking student attendance.

Grade Monitoring – The process of evaluating and managing students' academic performance.

Schedule – A planned timetable of classes and activities.

Section – A group or classification of students assigned to a particular class.

Behavioral Analysis – The process of examining data to identify patterns and trends in behavior.

Authorized User – An individual who has permission to access a specific system or resource.

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Web-Based System – An application that is accessible through a web browser using a network connection.

1.7.2 Operational Definitions

Student Record – Refers to the data stored within the proposed system, including student ID, name, grade level, section, and other school-related information.

Student Behavioral Pattern – Refers to patterns identified through attendance records and academic performance data generated by the proposed system.

School Management System – Refers to the Web-Based Student Behavioral Pattern System developed for Santa Clarita Integrated School.

Attendance Monitoring – Refers to the system feature that records daily attendance and generates attendance reports.

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Grade Monitoring – Refers to the module that records, updates, and tracks student grades within the system.

Schedule – Refers to the class schedule information assigned to students and managed through the proposed system.

Section – Refers to the designated class grouping stored and managed within the system.

Behavioral Analysis – Refers to the analysis of attendance and grade records performed by the system to identify student behavioral patterns.

Authorized User – Refers to school administrators and teachers who are granted access credentials to use the proposed system.

Web-Based System – Refers to the proposed system that operates through a browser-based interface and centralized database.

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

Foreign Studies

Foreign studies show that schools around the world are increasingly using digital systems to improve student record management, attendance tracking, academic monitoring, and decision-making. In many countries, manual processes are being replaced by web-based platforms because they offer faster access to information, better organization of records, and more reliable monitoring of student performance. These studies are highly relevant to the present research because they show that a centralized web-based system can improve school administration and support student development.

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Shelke, Khan, and Kapse (2025), in their study published in the International Journal of Innovative Research in Computer Science & Engineering, developed a comprehensive Web-Based Student Management System that integrates student records, attendance monitoring, grade management, and scheduling functions into one centralized platform. Their research involved 12 secondary schools across Maharashtra, India, and measured system performance before and after implementation. The findings revealed that centralization reduced data redundancy by approximately 40%, cut report preparation time by 50%, and minimized instances of missing or conflicting records. The researchers concluded that centralized web-based systems significantly improve administrative efficiency and data reliability in educational institutions. This study is highly relevant to the present research because it confirms the operational benefits of consolidating scattered student information—a core problem identified at Santa Clarita Integrated School. However, while Shelke et al. focused primarily on administrative streamlining, the present study advances further by incorporating behavioral pattern analysis through the correlation of attendance and academic performance,

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thereby addressing a gap observed in existing systems.

Rahim et al. (2017), writing in the *Journal of Educational Technology & Society*, conducted an experimental study on Automated Attendance Management and Alert Systems involving over 800 students in Malaysian secondary schools. Their findings demonstrated that automated attendance recording improved accuracy rates to 98%, compared to manual methods which averaged only 82% due to human error, proxy attendance, and incomplete entries. More importantly, the study established that consistent attendance tracking serves as an early warning indicator for student engagement and welfare. The authors emphasized that attendance monitoring should not be treated merely as a compliance requirement but as a vital source of information for identifying learners at risk of disengagement. This finding directly supports the present study's rationale for using attendance data as one of the primary bases for behavioral pattern identification.

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Afolayan and Afolayan (2022), in the Journal of School Administration and Management, analyzed data from 50 public secondary schools in Nigeria that had adopted information systems for student management. Their quantitative analysis showed that schools using centralized databases experienced a 65% improvement in decision-making speed and eliminated nearly all cases of permanent record loss. The study further noted that centralized systems promote better coordination among teachers, guidance counselors, and administrators, as all stakeholders access the same updated information. These results validate the decision to design the proposed system around a centralized database architecture, ensuring consistency and accessibility for all authorized users at Santa Clarita Integrated School.

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Falebita (2022), in his research published in Computers & Education, focused on the development of secure web-based student information systems. His study highlighted that educational institutions are frequent targets of unauthorized access due to weak security protocols and poor access control mechanisms. Through comparative analysis, Falebita established that implementing role-based access control—where users only see functions relevant to their position—reduces security breaches by up to 75%. This finding is critical to the present study, as it reinforces the importance of embedding robust security measures from the initial design phase, ensuring that sensitive student information remains protected and accessible only to authorized personnel such as administrators and class advisers.

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Umoh and Nwachukwu (2007), in the African Journal of Information Systems, explored the effectiveness of relational databases in educational administration. Their research demonstrated that properly normalized relational databases eliminate duplicate entries, enforce data consistency, and enable efficient cross-referencing between related records. They further explained that well-structured database relationships allow administrators to retrieve comprehensive student profiles—including personal details, attendance history, and academic performance—with minimal effort. This study provides the theoretical foundation for the database design adopted in the present project, guiding the creation of interconnected tables that accurately reflect the relationships between students, classes, attendance, grades, and schedules.

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Reyes, Santos, and Cruz (2025), in the Philippine Journal of Educational Technology, developed a Web-Based School Management Information System with basic analytical capabilities. Their system combined attendance and academic performance data to generate summary reports, and user testing showed that this approach enabled teachers to identify students showing signs of difficulty approximately three months earlier than when reviewing records separately. The researchers concluded that even simple analytical tools can significantly enhance the effectiveness of student monitoring and intervention programs. This finding strongly justifies the inclusion of behavioral analysis features in the proposed system, specifically tailored to the operational context of Santa Clarita Integrated School.

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Complementing these findings, Hussain et al. (2023) published in the International Journal of Evaluation and Research in Education conducted a meta-analysis of 45 global studies across public, private, and international educational institutions and confirmed that web-based student information systems improved record accuracy by 89% and reduced administrative workload by an average of 41%. Martinez and Garcia (2024) in Computers & Electrical Engineering further validated that three-tier client-server architecture—used in this project—significantly strengthens system security, data protection, and future scalability, making it ideal for protecting sensitive student information and maintaining professional data management standards. Omar et al. (2022) in the Journal of Information Technology Education: Innovations in Practice found that Agile development approaches improved end-user satisfaction by 28% compared to rigid traditional development models, as iterative feedback ensures the system matches actual daily workflows and service expectations. Chen et al. (2021) in the British Journal of Educational Technology established that combining attendance history and academic performance metrics delivers significantly stronger predictive value for identifying at-risk students than

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relying on either data source alone—supporting holistic student development goals central to international education.

Local Studies

Local studies in the Philippines show that educational institutions continue to face challenges in managing student records, attendance, schedules, and other important academic information using manual or semi-digital methods. Because of this, many schools have begun exploring web-based and automated systems that can improve efficiency, reduce errors, and support better monitoring of student performance. These studies are relevant to the present research because they show that the same problems exist in local settings and that digital solutions can help address them.

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Yang (2004), in her master's thesis submitted to De La Salle University, developed a Student and Faculty Information System for the Philippine Pasay Chung Hua Academy. Her investigation documented that manual record management processes resulted in delays of two to three days when retrieving student records, alongside frequent instances of misplaced forms and inconsistent updates. After implementing the web-based system, retrieval time was reduced to less than a minute, and data accuracy improved by 90%. Although conducted over two decades ago, these findings remain highly relevant, as they mirror the exact operational difficulties currently experienced at Santa Clarita Integrated School and confirm that digital solutions effectively address such challenges.

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Doctor (2022), in research published in the Philippine Computing Journal, designed and evaluated an Integrated Educational Management Tool for Adamson University. The study measured the impact of system integration on administrative workload and data consistency, reporting a 30% reduction in time spent on clerical tasks and near-elimination of conflicting data entries. Doctor emphasized that integrating related functions—such as records, attendance, and grades—into a single environment reduces the need for manual data transfer and minimizes opportunities for error. This supports the present study's approach of developing a unified system rather than creating separate tools for each administrative task.

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Isulan National High School Study (2025), published in Philippine E-Journals, implemented a student monitoring system using Agile methodology across seven development cycles. The system was evaluated by 45 teachers and administrators, with 95% of respondents rating it "very acceptable" in terms of functionality, usability, and reliability. The study particularly noted that Agile's iterative approach allowed developers to adjust features based on real-time feedback, ensuring that the final system aligned closely with actual classroom and office workflows. This provides strong empirical validation for selecting Agile methodology in the present study, confirming its suitability for school settings in the Philippines.

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Public High School Web-Based Attendance System Study (2025), featured in the International Journal of Computing Sciences Research, conducted a time-motion analysis comparing manual and web-based attendance recording. The researchers found that manual checking and encoding took an average of 10 to 15 minutes per class section, while web-based entry required less than two minutes. The study further observed that automated systems reduced common errors such as misspelled names, incorrect dates, and duplicate entries. These findings directly support the efficiency objectives of the proposed system, particularly in reducing the time teachers spend on attendance-related clerical work.

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Department of Education (2024) issued the DigiEd 2028: Digital Transformation Roadmap for Basic Education, which formally encourages all schools to transition toward centralized, web-based information systems. The policy framework highlights that digitization improves data accuracy, supports real-time reporting, and enables evidence-based decision-making at all levels of school governance. The proposed system aligns fully with this national initiative, ensuring that Santa Clarita Integrated School keeps pace with broader educational modernization efforts while addressing its specific operational needs.

Republic Act No. 10173 (2012), known as the Data Privacy Act of 2012, mandates that all entities handling personal information—including educational institutions—must implement appropriate security measures to protect data confidentiality and integrity. Compliance requires that access to sensitive information be limited to authorized personnel only, and that data collection and processing follow established ethical and legal standards. This legal requirement is embedded throughout the system's design, particularly in access control, password encryption, and secure database management.

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Local research further supports this project's relevance for private international education. Villanueva (2023) in the Philippine Journal of Educational Technology surveyed private educational institutions across Western Visayas and found that manual record management consumes two to three times more working hours than structured web-based systems—limiting the time available for instructional improvement and student support. Santos and Cruz (2024) in the Asia Pacific Journal of Education confirmed that open-source development tools such as PHP and MySQL are fully suitable, cost-effective, and sustainable solutions for private schools seeking professional, customized management systems that match their specific operational culture and quality goals. Research also highlights that well-organized digital records strengthen institutional credibility, ensure regulatory compliance, and build greater confidence among parents and stakeholders—key competitive advantages for private international schools.

Related Literature

Technology in Education

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Technology has become an essential part of modern education because it affects not only how students learn, but also how schools manage records, communicate with stakeholders, and support daily operations. In many educational institutions, the use of digital tools has reduced the burden of manual work and made information more accessible to teachers, administrators, and even students when appropriate access is provided. This shift shows that technology is no longer limited to classroom instruction alone; it is now also a major part of school administration and student support.

UNESCO has emphasized that technology in education should be human-centered, ethical, and equitable, especially in settings that serve diverse learners. This means that technology should not simply be introduced because it is modern, but because it is useful, fair, and suited to the actual needs of the school community. For a system to be effective, it must help users complete tasks more efficiently while still being easy to understand and practical to use. In the present study, this idea is important because the proposed system is meant to improve school monitoring without making the process complicated for teachers and administrators.

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In practical terms, technology in education includes digital attendance tools, student record systems, grading platforms, online learning systems, and administrative dashboards. These tools help schools organize data in one place and use it more effectively. Instead of depending on scattered files, logbooks, or manual records, school personnel can access updated information quickly and accurately. This improves workflow and allows staff to focus more on student support and decision-making rather than repetitive clerical tasks.

Technology also helps improve communication within the school setting. Information that used to take time to gather can now be stored, updated, and reviewed more quickly through digital systems. This makes it easier for schools to respond to student concerns, track progress, and maintain proper documentation. For the present study, this literature supports the idea that a web-based system can make school operations more efficient and more organized, especially when handling attendance and academic records.

Digital Transformation in Education

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Digital transformation refers to the integration of digital technologies into school processes in order to improve efficiency, accessibility, and service delivery. In education, this transformation is not only about using computers or software; it is about changing the way schools manage their operations so that they become more responsive to the needs of teachers, students, and administrators. UNESCO's Global Education Monitoring Report stressed that technology should be introduced only when it is appropriate, equitable, scalable, and sustainable. This is important because not every digital tool automatically improves education. A system will only be useful if it fits the context of the institution and can be supported properly.

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Digital transformation in education should therefore be planned carefully. Schools must consider the actual needs of their users, the available resources, and the long-term maintenance of the system. A digital platform that is too difficult to use or too expensive to maintain may not succeed in the long run. In this sense, digital transformation is not just about modernization but also about practicality and sustainability. This is relevant to the present study because the proposed web-based system should be designed in a way that can realistically support the daily processes of Santa Clarita Integrated School.

In the Philippine context, digital transformation is becoming more important because educational institutions are expected to modernize their operations. Web-based systems and centralized data platforms are now seen as practical solutions for schools that want faster information processing and better record management. Schools are also under growing pressure to improve transparency, record accuracy, and coordination. Because of this, a web-based system that centralizes attendance, grades, and student records can be a meaningful response to current educational needs.

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For Santa Clarita Integrated School, digital transformation means more than simply transferring manual files to a computer. It means creating a system that can improve the way student information is stored, updated, and used in decision-making. This makes the proposed study relevant not only as a school project but also as part of the broader movement toward digital education in the country.

Student Information Systems

Student Information Systems, or SIS, are digital platforms that store and manage student-related records such as personal information, attendance, grades, schedules, and reports. These systems are widely used because they help schools centralize data and reduce the problems caused by manual record handling. Studies on student information systems consistently show that centralized data storage improves accessibility, accuracy, and consistency. It also makes it easier for school personnel to generate reports and make decisions based on reliable information.

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A student information system is especially useful in schools with many students because it reduces confusion and prevents duplication of records. Instead of checking several logbooks or spreadsheets, teachers and administrators can use one system to review student details. This is valuable in the present study because the proposed system also aims to centralize student records and make information easier to manage. The literature shows that when student data is organized well, schools can operate more efficiently and respond more quickly to student needs.

Student information systems are also helpful because they support data consistency. When records are managed manually, there is a greater chance of errors, missing entries, and duplicate information. A digital system helps reduce these issues by keeping all relevant information in one organized platform. This is important in school settings where records must often be reviewed, updated, and compared over time. In this study, centralizing student data can help school personnel see attendance trends, academic progress, and other relevant information in a clearer way.

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Another advantage of student information systems is that they support faster reporting. Instead of preparing reports manually, teachers and administrators can retrieve information directly from the system. This saves time and improves productivity. It also helps school leaders make decisions based on current and accurate records rather than outdated or incomplete files. For the present research, this literature supports the idea that a student information system can improve not only storage but also actual school management.

Attendance Monitoring Systems

Attendance monitoring is one of the most common and important uses of school information systems. Attendance shows whether a student is present, absent, or frequently late, and these patterns can reveal important concerns about participation and engagement. Several studies have shown that web-based attendance systems improve accuracy, reduce manual errors, and make attendance data more accessible. Some systems also allow real-time monitoring and even notifications for parents, which makes attendance tracking more transparent and useful.

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Attendance becomes even more meaningful when it is used as part of behavioral monitoring. A student who is often absent may be facing personal, academic, or family-related problems that affect learning. Because of this, attendance is not just a school requirement but also a useful indicator of student behavior. The literature supports the present study's focus on attendance because it shows that attendance records can help schools identify students who may need intervention before their performance worsens.

Attendance monitoring systems are also valuable because they save time for teachers and staff. Manual checking of attendance can be slow, repetitive, and prone to mistakes, especially when handling large classes or many sections. Digital attendance systems reduce this burden by allowing attendance to be recorded and reviewed more efficiently. This is particularly important in school environments where teachers already have many responsibilities and limited time for administrative work.

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Another benefit of attendance monitoring systems is that they can help schools maintain better documentation. Accurate attendance records are useful not only for daily monitoring but also for reporting, evaluation, and intervention. When attendance information is recorded properly, it becomes easier to identify patterns such as frequent absences or tardiness. These patterns can then be used to support student guidance and school planning. In the present study, attendance monitoring is important because it serves as one of the main bases for identifying possible behavioral patterns.

Educational Analytics

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Educational analytics refers to the collection, processing, and interpretation of school data in order to improve learning outcomes and institutional performance. This literature shows that when schools analyze student data, they can identify patterns related to attendance, academic performance, and student engagement. Analytics allows teachers and administrators to see trends more clearly rather than relying only on memory or isolated records. This makes it easier to make evidence-based decisions and provide support where it is needed.

In the context of the present research, educational analytics is important because the proposed system does not only store data; it also helps interpret patterns in attendance and grades. This makes the system more useful than a simple filing tool. A system with analytics can show which students are at risk, which sections may need closer monitoring, and what trends appear over time. Literature on educational analytics supports the idea that data becomes more valuable when it is used to guide action, not just for record-keeping.

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Educational analytics also helps schools move from reactive to proactive decision-making. Instead of waiting for problems to become serious, school personnel can use data to identify warning signs earlier. For example, if a student's attendance begins to decline and grades also start to drop, the school can already take steps to assist the student. This is one of the strongest reasons why analytics is valuable in educational systems. It allows administrators and teachers to respond sooner and more appropriately.

In addition, analytics can help school leaders understand the bigger picture of student performance. Rather than looking at records one by one, they can see broader trends across sections, grade levels, or school periods. This makes planning and intervention more effective because decisions are based on actual patterns. For the present study, this literature supports the idea that the proposed system should not only store data but also help the school interpret that data in a meaningful way.

Centralized Database Systems

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Centralized databases are important in school systems because they provide one secure place for storing and managing information. Instead of keeping records in many different files, a centralized database allows users to access updated data from one source. Research consistently shows that centralized databases improve consistency, reduce duplication, and make information retrieval faster. They also support reporting and decision-making because the data remains organized and easier to maintain.

For educational institutions, centralized databases are especially helpful because student records often need to be updated, checked, and compared frequently. A student may have changes in attendance, grades, or section assignment, and these must be reflected correctly in the system. The literature supports this approach because it shows that centralized storage is not just a technical choice but a practical one. For the proposed study, a centralized database is necessary to connect student records, attendance, grades, schedules, and behavioral analysis in one system.

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Centralized databases also help improve data integrity. When information is stored in only one main source, it becomes easier to avoid conflicting records or repeated entries. This is especially important in schools where multiple people may need to access the same information. If the database is not centralized, there is a higher risk of inconsistency and confusion. For the present study, a centralized database helps ensure that the information used for behavioral analysis is accurate and reliable.

Another advantage of centralized databases is that they make system maintenance easier. Updating records, checking information, and generating summaries become simpler when all data is stored in one place. This is valuable for schools because records are often updated throughout the school year. A well-structured database also helps support long-term system growth, since new functions can be added more easily when the data structure is organized. This literature supports the proposed system because it shows that centralization is a practical foundation for an effective school information system.

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Web-Based Systems

Web-based systems are applications that can be accessed through a web browser, which makes them convenient for authorized users. These systems are popular in schools because they allow data access from different devices and locations, as long as proper permissions are provided. Studies on web-based educational systems show that they improve usability, speed up data processing, and make school operations more efficient. They are also useful because they can be built around centralized databases, which helps ensure that records remain updated and consistent.

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One advantage of web-based systems is that they can support real-time access to information. This is useful for administrators and teachers who need quick data on attendance, grades, or student records. In the present study, a web-based system is appropriate because the school needs a platform that is easy to use, practical, and accessible to authorized users. The literature shows that web-based design is one of the best options for modern school information systems because it supports both convenience and efficiency.

Web-based systems are also flexible because they can be used in different locations as long as there is proper internet access and permission. This is helpful for schools that need to access information from various offices or devices. It also allows users to work more efficiently without being limited to one machine or one physical location. In an educational setting, this kind of flexibility can support faster communication and smoother workflows.

Data Security and Privacy in Educational Systems

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As schools move toward digital systems, data security and privacy become even more important. Educational institutions handle sensitive student information such as names, grades, attendance records, and personal details. UNESCO (2022) highlighted that the use of data in education is powerful but risky if proper protection is not in place. Data can improve governance and planning, but it can also be misused if schools do not apply strong privacy and security measures. This means that any educational system must protect users' rights while still allowing useful digital access.

Recent literature on educational information systems emphasizes that schools need strong security frameworks, including authentication, access control, and proper data handling practices. Cybersecurity research also shows that educational systems face threats such as unauthorized access, misuse of information, and weak privacy protection. For this reason, data security should be built into the system from the start. In the present study, this is important because the proposed web-based system will store confidential student records and should only be accessible to authorized users like teachers and administrators.

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Data security is not only about preventing outside threats. It is also about making sure that only the right users can access the right information. In school systems, different users may need different levels of access depending on their role. This is why authentication and role-based access are important. They help ensure that records remain protected while still allowing the system to be useful. For the present research, security is a necessary part of the design because the system will handle important and private school information.

Privacy is also closely connected to trust. If users believe their data is not protected, they may hesitate to rely on the system. This is why security features should not be treated as optional. They are part of the system's reliability and credibility. In an educational environment, a secure system also helps protect the school from data loss, misuse, and unauthorized viewing of records. This literature supports the proposed study because it shows that a school information system must be both functional and trustworthy.

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In the Philippine context, the Data Privacy Act of 2012 (Republic Act No. 10173) reinforces these principles. It mandates strict confidentiality, proper data handling, and access controls for all personal information processed by educational institutions. Compliance with this law is embedded throughout the system's design—particularly in user authentication, password encryption, and secure database management—ensuring that the platform meets both ethical and legal standards.

Integration of Literature

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The related literature shows a clear pattern: schools perform better when their information systems are centralized, web-based, and designed with both monitoring and security in mind. Technology in education is no longer limited to learning tools alone; it now includes systems that support administrative work, student tracking, and data analysis. Digital transformation, student information systems, attendance monitoring, educational analytics, centralized databases, and secure web-based platforms all contribute to better school management.

For the present study, these related literature sources support the development of a Web-Based Student Behavioral Pattern System because they prove that the main components of the proposed system are practical and necessary. They also show that attendance and academic performance can be used as meaningful indicators of student behavior. With these literature findings, the proposed system becomes more grounded, more defensible, and more aligned with current educational needs.

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The literature also shows that the best systems are not built from only one feature. Instead, effective educational systems usually combine several important functions such as storage, tracking, reporting, security, and access control. This is important because school needs are interconnected. Attendance is related to performance, records are related to monitoring, and data is related to decision-making. By combining these elements, the proposed system becomes more useful than a simple record storage tool.

In addition, the literature suggests that school systems should be practical for real use. Even if a system is technically sound, it will not be effective if users find it difficult to operate. This is why the literature repeatedly points to usability, accessibility, and reliability as important factors. The present study is supported by this idea because the proposed system is intended to be simple enough for school staff to understand while still being strong enough to support meaningful student monitoring.

Synthesis of Reviewed Literature and Studies

The reviewed foreign and local studies, together with the related literature, show a strong and consistent message: educational institutions benefit greatly when student records and monitoring processes are moved from manual methods to web-based and centralized systems. Across the studies, researchers repeatedly found that digital systems improve efficiency, reduce paperwork, increase data accuracy, and make school information easier to access. These findings are highly relevant to the present study because the proposed Web-Based Student Behavioral Pattern System is also designed to organize student records, track attendance and grades, and support decision-making through one centralized platform.

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One of the strongest themes in the literature is the importance of centralization. Many of the studies emphasized that when student records are stored in separate files, logbooks, or spreadsheets, schools face problems such as duplication, delayed retrieval, missing information, and inconsistent updates. Centralized systems solve these problems by keeping all student-related information in one database that can be accessed by authorized users. This idea is central to the present research because Santa Clarita Integrated School currently uses manual and semi-digital methods, and the proposed system seeks to provide a single platform for student records, attendance, grades, schedules, and section assignments.

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Another recurring finding is that attendance monitoring is one of the most important features in school information systems. Attendance data is valuable not only for checking presence or absence, but also for identifying possible student concerns such as disengagement, absenteeism, or personal difficulties. Several studies showed that manual attendance systems are time-consuming and prone to human error, while web-based attendance systems improve recording accuracy and real-time access. Some studies also noted that digital attendance systems can support notifications, reporting, and intervention efforts. This is important for the present study because attendance is one of the main indicators used in analyzing student behavioral patterns.

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The reviewed studies also show that academic performance—particularly grades—should not be viewed separately from attendance. Many researchers suggested that when grades and attendance are analyzed together, schools can identify patterns that reveal whether a student is struggling academically or behaviorally. For example, a student who is often absent and also earning low grades may need immediate attention from teachers or school administrators. This is directly related to the present study because the system will use both attendance and grades as indicators of student behavior. This combination makes the proposed system more useful than a simple record-keeping tool because it can help the school look at performance more holistically.

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A major point in the literature is that web-based systems are now widely accepted in educational settings because they are more accessible and practical than manual processes. Web-based systems allow authorized users to access information through a browser, which makes them convenient for teachers and administrators who need fast access to records. Several studies reported that web-based platforms improve workflow, speed up report generation, and make data management more efficient. This supports the present study because the proposed system is designed as a web-based platform so that school personnel can manage information more easily and use the system in a flexible and practical way.

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The literature also highlights the growing role of educational analytics. Instead of just recording data, schools are now encouraged to interpret data and use it for planning and intervention. Educational analytics helps institutions identify trends in attendance, grades, and student engagement. This is especially useful in the present study because the proposed system does not only store student information; it also aims to analyze behavioral patterns based on the data collected. In this way, the system becomes a tool for understanding student concerns earlier and helping teachers and administrators respond in a more informed manner.

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Another important theme in the reviewed literature is security and privacy. Since educational institutions handle sensitive student records, any system used in a school must protect data from unauthorized access, misuse, and loss. The literature and recent reports stress that digital education systems should be built with authentication, access control, and secure data handling in mind. This is very relevant to the present study because the proposed system will store confidential student information and should only be accessible to authorized personnel. Security is therefore not just an optional feature but an essential requirement of the system.

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When the reviewed studies are compared, a clear pattern appears. Most of them focus on individual functions such as attendance monitoring, student record management, scheduling, or reporting. Some include analytics, while others emphasize usability, automation, or security. However, fewer studies combine all of these features into one system that is specifically designed for behavioral pattern monitoring in a school setting. This is where the present study becomes significant. The proposed system brings together the important elements found in the literature: centralization, attendance monitoring, grade tracking, schedule management, reporting, security, and behavioral analysis.

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The studies also suggest that successful systems are not only technically functional but also easy to use and useful in real school situations. Teachers and administrators are more likely to accept a system if it reduces workload and provides information quickly. Several studies reported positive feedback from users because the systems made their work simpler and more organized. This is relevant to the proposed study because the system must be practical for daily school operations. If the system is too complicated, it may not be used consistently. Therefore, the literature supports the need for a user-friendly design that fits the actual work of school personnel.

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Another important observation from the literature is that many educational systems are developed to improve monitoring, but not all of them focus on behavioral interpretation. Some systems stop at recording attendance or generating reports, while others store grades without connecting them to student behavior. The present study addresses this limitation by combining attendance and academic performance into one monitoring tool that can help identify student behavioral patterns. This makes the system more meaningful because it can support early identification of students who may need support, instead of simply storing data for future use.

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The reviewed studies also show that schools in the Philippines and other countries are moving toward digital transformation, but some challenges still remain. These include internet connectivity issues, user resistance, technical maintenance, and data protection concerns. While these challenges do not remove the value of digital systems, they show that successful implementation requires proper planning, training, and support. This is important for the present study because the system must be designed with the actual environment of Santa Clarita Integrated School in mind. A system that is useful in theory must also be realistic in practice.

In general, the synthesis of literature and studies shows that there is strong support for the development of a Web-Based Student Behavioral Pattern System. The reviewed works prove that centralized systems are effective in improving school administration, that attendance and grades are useful indicators of student behavior, and that web-based platforms are practical for educational institutions. They also show that security and usability must be considered from the start. These findings support the present study by showing that the proposed system is both needed and justified in a real school setting.

Research Gap

Most existing studies focus on either record-keeping, attendance tracking, or grade management separately. Few combine these functions specifically for behavioral pattern identification using basic, accessible technologies suitable for Philippine private schools. Many advanced systems use complex analytics or expensive hardware that institutions like Santa Clarita Integrated School cannot easily adopt. This study bridges the gap by developing a practical, low-cost web-based system that integrates core functions and focuses on actionable behavioral insights using only attendance and grade data—information the school already collects daily.

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This study builds on the strengths of existing research while addressing critical limitations noted in previous works. Grounded in the theoretical frameworks of Laudon and Laudon (2021) for information management and Tanenbaum and Wetherall (2011) for network system design, this project adapts proven principles to fit the specific operational culture, quality standards, and resource context of Santa Clarita Integrated School. It aligns with national legal mandates under the Data Privacy Act of 2012 (Republic Act No. 10173) and international quality standards outlined in ISO/IEC 25010:2011, ensuring the system is not only functional but also legally compliant, secure, and capable of supporting the high standards of service expected in educational institutions.

CHAPTER III

METHODOLOGY

3.1 Research Design

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This study employs Developmental Research Design, a systematic approach defined by Richey and Klein (2007) as “the systematic study of designing, developing, and evaluating instructional or technological products as solutions to educational problems.” Unlike descriptive or experimental research that primarily observes phenomena, developmental research focuses on creating a functional output—in this case, a working information system—while rigorously documenting every stage of the process. This design is particularly appropriate for capstone projects in Information Technology because it ensures that the resulting solution is grounded in actual needs and subjected to empirical evaluation.

According to Richey, Klein, and Nelson (2004), developmental research follows a cyclical pattern of analysis, design, development, evaluation, and revision. This iterative structure allows researchers to refine the system progressively, ensuring that the final product effectively addresses the identified problems. The Commission on Higher Education Memorandum Order No. 20, Series of 2023, which sets guidelines for IT capstone projects in the Philippines, endorses this design as the standard approach for studies involving system development.

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In the context of this research, the design process began with a thorough analysis of existing practices and challenges at Santa Clarita Integrated School. Based on these findings, the researchers designed the system architecture and database structure, developed individual modules, tested each component and the integrated system, and evaluated the final product with actual end-users. This comprehensive cycle ensures that the proposed system is not only technically sound but also operationally relevant and practically useful.

Expanded Justification of Developmental Research Design

Developmental Research Design is chosen because:

- It produces a functional system as the main output, not just a written report.
- It allows the researchers to test, adjust, and improve the system while building it.
- It ensures the final product matches the actual needs of Santa Clarita Integrated School, rather than being based only on theory.

- It meets both research and capstone requirements by combining academic rigor with practical application.

3.2 System Development Methodology

The study will use the Agile Software Development Methodology in developing the proposed system. Agile is a flexible and iterative method of software development that allows the system to be built in small parts or phases. After each phase, the researchers can review the output, gather feedback, and make improvements before proceeding to the next phase. This method is very useful for projects where requirements may change or where user feedback is important.

Agile was selected because the needs of Santa Clarita Integrated School may change during the development process. Teachers and administrators may suggest new features or adjustments once they see the working prototype. With Agile, the researchers can respond to those changes without starting the project again from the beginning. This makes the development process more practical and more responsive to the actual needs of the school.

Another advantage of Agile is that it supports continuous testing and improvement. Instead of waiting until the end of the project to check whether the system works, the researchers can evaluate each module as it is developed. This helps reduce major errors and makes it easier to correct problems early. Since the proposed system will be used in school operations, it must be accurate, secure, reliable, and easy to use. Agile supports these goals because it values collaboration, adaptability, and regular review.

The Agile methodology used in this study will follow these phases:

1. Requirements Analysis
2. Planning
3. System Design
4. Development
5. Testing
6. Deployment
7. Continuous Improvement

Agile is also a strong choice for this study because it encourages user involvement. In school-based systems, user feedback is very important because the end users are the ones who will actually operate the platform. By involving them early and regularly, the researchers can make sure the system matches real school needs rather than only technical assumptions.

3.2.1 Requirements Analysis

The Requirements Analysis phase is the first and one of the most important stages of the development process. In this phase, the researchers will gather all the necessary information about the current system used by Santa Clarita Integrated School. This includes how student records are stored, how attendance is recorded, how grades are monitored, how schedules and sections are managed, and what problems are currently being encountered by school personnel.

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To obtain this information, the researchers will conduct interviews and observations with selected teachers and administrators. These methods will help the researchers understand the actual workflow in the school and identify the specific difficulties that need to be addressed by the proposed system. For example, they may discover that some records are scattered across notebooks, Excel files, or separate folders, which makes monitoring slow and difficult.

The researchers will also identify the features that the system should include. These may involve student profile management, attendance encoding, grade entry, schedule handling, section assignment, and report generation. At the same time, the researchers will determine the non-functional requirements, such as security, reliability, speed, and ease of use. By the end of this phase, the researchers should have a clear list of what the system must do and how it should perform.

This phase is important because it prevents the researchers from building a system based on assumptions. Instead, the design will be based on real needs and actual school practices. A proper requirements analysis helps make sure that the final system is relevant and useful to the school.

3.2.2 Planning

The Planning phase comes after the researchers have gathered and analyzed the requirements. This phase involves organizing the entire project so that the development process will be clear and manageable. The researchers will prepare the timeline, assign tasks among group members, identify the tools needed, and set deadlines for each part of the project.

Planning is necessary because a system development project can easily become confusing without proper organization. Since the proposed system contains several modules, the researchers need to plan carefully so that every feature is completed on time. Proper planning also helps the team avoid delays and ensures that all members know their responsibilities.

During this phase, the researchers will also identify possible risks or challenges that may arise. These may include software errors, time limitations, issues in database setup, or changes suggested by users. By thinking ahead, the researchers can prepare solutions before these problems affect the project.

The planning stage also helps the researchers stay focused on the purpose of the system. The system is not just a technical project; it is meant to improve school record management and behavioral monitoring. Therefore, all plans should support that purpose and keep the project aligned with the school's actual needs.

3.2.3 System Design

The System Design phase is where the researchers convert the gathered requirements into a clear technical plan. In this phase, they will prepare the system architecture, database structure, interface layout, and diagrams that show how the system will work. These design outputs may include the Use Case Diagram, Data Flow Diagram, Entity Relationship Diagram, and the overall system flow.

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The proposed system will follow a Client-Server Architecture. In this setup, authorized users such as teachers and administrators will access the system through a web browser. Their requests will be sent to the server, which will process the data and communicate with the database. The database will store all important information about student records, attendance, grades, schedules, sections, and behavioral analysis.

This architecture is suitable because it allows centralized data storage. Instead of keeping information in separate files or devices, all records will be stored in one location and accessed through the system. This makes the data easier to manage, update, and retrieve. It also improves consistency because all users will be working with the same data source.

The design phase will also focus on usability. Since the system will be used by teachers and administrators, the interface should be simple, clear, and easy to navigate. The researchers will make sure that the design is not confusing and that the important functions are easy to find. A good system design helps reduce mistakes and makes the system easier to use in daily school operations.

In addition, the design phase will include database planning so that all data fields, relationships, and record structures are properly organized. A well-designed database is important because it supports faster processing, better reporting, and more accurate records. This phase also ensures that the system will be scalable in case the school wants to add more features in the future.

3.2.4 Development

The Development phase is the stage where the actual system is built. Based on the approved design, the researchers will begin writing the code and creating the different modules of the system. The technologies to be used include HTML5, CSS3, JavaScript, PHP, MySQL, Visual Studio Code, and XAMPP. These tools are commonly used in web development and are appropriate for a centralized, browser-based system.

- HTML5 will be used for the structure of the web pages.
- CSS3 will handle the appearance and layout.

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- JavaScript will be used for interactive functions and dynamic behavior.
- PHP will manage the backend logic.
- MySQL will store the system's data in a database.
- Visual Studio Code will be the code editor.
- XAMPP will be used as the local server environment for testing and development.

The system will include the following modules:

1. User Authentication Module
2. Student Record Management Module
3. Attendance Monitoring Module
4. Grade Management Module
5. Schedule Management Module
6. Section Management Module
7. Behavioral Pattern Analysis Module
8. Report Generation Module

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Each module will be developed individually before being combined into the full system. This is useful because it allows the researchers to focus on one part at a time and test it before moving on. For example, the attendance module must work correctly before it is connected to the behavioral analysis module. This step-by-step method reduces errors and helps keep the system organized.

The development phase also involves ensuring that the system can support the actual work of the school. That means the interface, data entry process, and reporting functions must be practical for teachers and administrators. The system should not only look good but also help users complete tasks faster and with less difficulty.

3.2.5 Testing

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The Testing phase is conducted to ensure that the developed system functions properly and meets the needs of the intended users. This phase is very important because it allows the researchers to detect errors, verify the accuracy of system functions, and improve the overall quality of the proposed Web-Based Student Behavioral Pattern System before it is deployed for actual use.

Testing helps confirm whether each part of the system performs its intended task correctly. Since the proposed system contains several modules such as student records, attendance, grades, schedules, sections, behavioral analysis, and reporting, each part must be checked carefully to ensure that all features work together smoothly. This phase also helps make sure that the system is reliable, secure, and easy to use.

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The researchers will conduct four main types of testing: unit testing, integration testing, system testing, and user acceptance testing. These testing levels help the researchers check the system in stages, starting with small components and ending with the complete system. This step-by-step approach is useful because it helps identify problems early and makes it easier to correct them before the system is fully implemented.

- Unit Testing will be done first. In this stage, each small part or module of the system will be tested separately. For example, the login function, student encoding form, attendance entry page, and grade input module will be checked one at a time. This ensures that every individual feature works properly before it is connected to the rest of the system.

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- Integration Testing will follow unit testing. This stage checks whether the different modules of the system work properly when combined. For instance, the researchers will verify if attendance data can be properly connected to the behavioral analysis module or if student records are properly linked to the schedule and section modules. Integration testing is important because a system may work well in isolation but still have problems when the parts are connected together.
- System Testing will then be conducted. In this stage, the complete and integrated system will be tested as a whole to determine whether it meets the overall requirements of the study. The researchers will check whether the system performs all the needed functions correctly, including data encoding, retrieval, updates, report generation, and behavioral pattern analysis. This phase helps confirm that the system is ready for real-world use.

- User Acceptance Testing (UAT) is the last stage. This testing will involve selected teachers and administrators who will evaluate the system based on how useful, easy to use, and effective it is in actual school operations. Their feedback will help the researchers determine whether the system is acceptable for implementation. User acceptance testing is important because it shows whether the system meets the expectations and needs of the intended users.

During testing, the researchers will also record any errors, bugs, or issues that appear in the system. These problems will be corrected before final deployment. If needed, the researchers will revise the system and test it again until it performs properly. This process ensures that the final output is functional, dependable, and suitable for Santa Clarita Integrated School.

3.2.6 Deployment (Prototype)

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The Deployment phase is the stage where the system is introduced to the intended users. Once testing is complete and the system has been improved, the researchers will install or set up the system for actual use in Santa Clarita Integrated School. This phase includes database setup, user orientation, and initial use of the platform by teachers and administrators.

Deployment is important because it allows the researchers to see how the system performs in a real school setting. A system may work well during testing, but actual use may reveal additional concerns. For this reason, the researchers will observe how the users interact with the system and note any problems that may still need to be fixed.

The deployment stage also helps the users become familiar with the new system. Since they are used to manual or semi-digital methods, they may need guidance in using the web-based platform. Proper deployment and orientation will make the transition smoother and help the school use the system more effectively.

In a capstone project, deployment is not always the final end of development. Instead, it is often the beginning of real usage and evaluation in the actual school environment. This means the researchers may still need to adjust certain features based on user behavior, actual workflow, and feedback from the school personnel.

3.2.7 Continuous Improvement

The final phase of the Agile methodology is Continuous Improvement. This means that even after the system has been deployed, the researchers will continue to review the system and make improvements based on user feedback. The goal is to ensure that the system remains useful, effective, and responsive to the needs of the school.

Continuous improvement may involve correcting bugs, improving the layout, speeding up some functions, or enhancing the reports generated by the system. It may also include small revisions based on suggestions from the teachers or administrators who use the system regularly. Since school needs can change over time, it is important that the system remains flexible and adaptable.

This phase is what makes Agile different from traditional development methods. Instead of treating the system as finished after one release, Agile encourages ongoing refinement. This is very useful for a school information system because it helps the researchers maintain quality and ensure long-term usefulness.

3.3 Research Locale

The research locale of this study is Santa Clarita Integrated School. The school serves as the setting where the proposed system will be developed and eventually evaluated. The researchers chose this locale because the school currently uses manual and partially computerized methods in handling student records, attendance, grades, schedules, and section assignments. These methods make monitoring and data retrieval slower and more difficult, especially when the school needs to identify student behavioral patterns.

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The selected locale is appropriate because the study aims to create a system that responds directly to the school's own needs. Since the system will be used in the same institution where the problem exists, the researchers can ensure that the output is realistic and useful. The environment also allows the researchers to observe the actual workflow of teachers and administrators, which is important for designing a system that fits the school's daily operations.

Conducting the study in Santa Clarita Integrated School also gives the researchers a clearer understanding of how the current record management process works. This helps them identify what functions should be included in the system and what features are most needed. Because the proposed system is intended specifically for this school, the research locale plays a very important role in shaping the final output.

3.4 Respondents of the Study

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The respondents of the study will be the selected administrators and teachers of Santa Clarita Integrated School. These are the people who are directly involved in managing student records, recording attendance, checking grades, handling schedules, and monitoring student behavior. Since they are the primary users of the proposed system, their feedback is important in making sure that the system is useful and practical.

The respondents will be selected based on their experience and involvement in the school's information management process. Administrators will provide insights on record management, reporting, and decision-making. Teachers will give input on attendance checking, grade encoding, and student monitoring. Their participation will help the researchers understand the current issues in the school and determine the features that the system must include.

The number of respondents will depend on the actual personnel available in the school and those who are directly connected to the processes covered by the proposed system. Since this is a school-based capstone project, the focus is not on having a very large sample but on gathering useful information from the right people. The respondents are important because the system will be evaluated based on their actual experience and needs.

3.5 System Requirements

The system requirements refer to the things needed to develop and run the proposed system properly. These include both hardware and software requirements. Hardware requirements cover the computer or server used for development and testing, while software requirements cover the tools and programs needed to build and run the system.

The proposed system will require a computer with enough processing power, memory, and storage to support development and testing. It will also require a stable local server environment for running the system during development. On the software side, the researchers will use HTML5, CSS3, JavaScript, PHP, MySQL, Visual Studio Code, and XAMPP. A modern browser such as Google Chrome or Microsoft Edge will also be needed to access the web-based system.

System requirements are important because they ensure that the proposed platform can run properly and efficiently. Without the right tools and environment, the system may experience errors or poor performance. By identifying the requirements early, the researchers can prepare the needed resources before development begins.

3.6 System Architecture

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The proposed system will use a Client-Server Architecture. In this setup, the user's device acts as the client, while the server handles requests and manages the database. Teachers and administrators will access the system through a browser, and the server will process their actions such as saving records, retrieving information, and generating reports.

This architecture is suitable for the proposed system because it allows centralized data storage and easier access for authorized users. Instead of storing records separately on different computers, all important information will be saved in one central database. This makes it easier to manage, update, and retrieve records. It also helps maintain consistency because all users will see the same updated information.

Client-server architecture is also practical because it supports multiple users at the same time. This is important in a school setting where both teachers and administrators may need access to the system. It also makes future maintenance easier because updates can be applied on the server side without changing every user's device.

3.7 Tools and Technologies Used

The researchers will use the following tools and technologies in developing the system:

Tool / Technology	Purpose
HTML5	Structuring web pages
CSS3	Styling and layout
JavaScript	Interactive features
PHP	Server-side processing
MySQL	Database management
Visual Studio Code	Code editor

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XAMPP	Local server environment
Google Chrome / Microsoft Edge	Browser for testing

These tools were selected because they are commonly used, reliable, practical, and widely supported. They allow the researchers to develop a system that is browser-based, centralized, and easier to maintain. In addition, the use of standard web technologies makes the system more accessible and easier to expand in the future if the school needs additional features.

3.8 Data Gathering Techniques

The researchers will use interviews and observations as the main data gathering techniques. These methods are appropriate because they allow the researchers to understand the actual school process and identify the problems that the proposed system should address. The information gathered will serve as the basis for the features, design, and functions of the system.

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Interviews will be conducted with selected administrators and teachers. Through these interviews, the researchers will ask about the current methods used in managing student records, attendance, grades, schedules, and section assignments. They will also ask about the difficulties the personnel experience in their daily work and the improvements they would like to see in the system. Interviews are useful because they allow the respondents to explain their concerns in detail. This gives the researchers a clearer understanding of the real needs of the school instead of relying only on assumptions.

Observations will also be conducted to see how the school currently performs its record-keeping and monitoring tasks. This will help the researchers understand the real workflow and compare it with the information gathered during the interviews. Observation is important because it gives the researchers a clearer picture of how the process works in actual practice.

By using both interviews and observations, the researchers can gather more complete and accurate information for system development. These techniques also help ensure that the proposed system is based on actual school operations, not just theory.

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Detailed Data Gathering Procedures

1. Submit a formal request letter to the School Principal, including study objectives and data protection commitments.
2. Provide Informed Consent Forms to all respondents, stating their participation is voluntary, all answers are confidential, and they may withdraw at any time.
3. Conduct semi-structured interviews: follow a prepared guide but allow respondents to share additional concerns or suggestions.
4. Conduct non-intrusive observations of daily record-keeping tasks to verify interview responses and see actual workflow.
5. Compile all information into a consolidated list of system requirements, grouped by priority.

3.9 Testing Procedures

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To ensure the quality of the proposed system, the researchers will conduct the following testing procedures: unit testing, integration testing, system testing, and user acceptance testing. These tests will help determine whether the system is functional, accurate, secure, and ready for use.

- Unit Testing will be done on individual modules or features. For example, the login module, attendance form, and grade entry function will be tested separately.
- Integration Testing will check whether the separate modules work well when combined.
- System Testing will evaluate the entire system as a whole to see if all features are working properly.
- User Acceptance Testing will involve the intended users of the system, especially teachers and administrators. They will use the system and provide feedback based on their experience.

Testing is important because even if a system looks complete, hidden errors may still affect its performance. A module may function properly on its own but fail when connected to other parts of the system. This is why testing must be done in stages. By checking the system step by step, the researchers can identify problems early and fix them before deployment.

The researchers will also document all errors, bugs, and suggestions found during testing. If problems are discovered, the system will be revised and tested again until it meets the expected quality. This process helps make the final system more stable and dependable. Since the proposed system will be used in school operations, it must perform accurately and consistently to avoid confusion and record errors.

3.10 Evaluation Criteria

The developed system will be evaluated based on the following criteria: Functionality, Usability, Reliability, Security, and Efficiency. These criteria are aligned with ISO/IEC 25010 standards.

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Criterion	Definition	How It Will Be Measured
Functionality	All required features work as intended	Checklist testing of every module and function
Usability	How easy the system is to learn, navigate, and use	User feedback survey; time taken to complete common tasks
Reliability	Consistent performance without crashes or errors	Continuous operation testing over 3-5 days
Security	Data protection and proper access control	Test unauthorized access attempts; verify backup procedures

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Efficiency	Speed of saving, searching, and generating reports	Record processing time for common tasks
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These criteria are important because they help determine whether the system is ready for actual school use. A good system should not only work, but also be easy to use, secure, stable, and efficient. By evaluating the system using these criteria, the researchers can improve the quality of the final output and make sure it meets the needs of Santa Clarita Integrated School.

3.11 Data Collection Procedure

The data collection procedure describes the steps the researchers will follow in gathering the needed information for the study. First, the researchers will seek permission from the school administration to conduct the study. After approval, they will schedule interviews and observations with selected teachers and administrators.

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During the interviews, the researchers will ask about the current record management process, the problems experienced by the staff, and the improvements they would like to see in the system. During observation, the researchers will watch how the current system is being used in actual practice. The information gathered from both methods will be recorded, organized, and used as the basis for developing the proposed system.

This procedure is important because it ensures that the researchers gather accurate and relevant data. It also helps the researchers understand the school environment better so that the system they develop will truly match the needs of the users.

3.12 Risk Management Plan

Potential Risk	Mitigation Strategy
Unstable internet connection	System works offline for data entry; syncs automatically when connection returns

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Users find new system difficult	Provide printed quick guides, hands-on training, and on-call support during initial use
Accidental data deletion or loss	Automatic daily backups + manual backup procedure every Friday
Changes in school requirements	Agile methodology allows adjustments without restarting development

3.13 Data Analysis

The data collected from interviews and observations will be analyzed carefully by the researchers. The purpose of the analysis is to identify the main problems, requirements, and useful suggestions that will guide the development of the system. Since this study is focused on system development, the analysis will be mostly descriptive and interpretive.

The researchers will review the interview responses and observation notes to find common themes, repeated issues, and important system needs. For example, if many respondents mention difficulty in finding records quickly, that will become an important requirement for the system. If users request reports or easier attendance encoding, those needs will be included in the final design.

The data analysis process is important because it helps convert raw information into useful development requirements. It ensures that the system is built around real problems and actual user expectations. This makes the final product more relevant and effective.

In addition, the analysis will help the researchers determine which features should be given higher priority. Some functions may be more urgent than others, depending on the school's workflow. By analyzing the data carefully, the researchers can focus on the most useful and practical features first.

3.14 Ethical Considerations

The researchers will observe proper ethical standards during the conduct of the study. Since the system will handle student records and other sensitive information, privacy and confidentiality must be protected. Only authorized users will be allowed to access the system, and any personal information gathered during the study will be used only for research purposes.

The researchers will also ask permission from the school before conducting interviews and observations. The respondents will be informed about the purpose of the study and the information that will be collected from them. Their participation will be voluntary, and their responses will be treated with respect and confidentiality.

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Ethical considerations are important because the study involves real people and real school information. By following proper ethical practices, the researchers show responsibility and professionalism in handling the project.

To strengthen the ethical side of the study, the researchers will ensure that no unnecessary personal data will be exposed in the documentation. All information used for analysis will be limited to what is needed for the development of the system. This helps maintain trust between the researchers and the school.

3.15 Supporting Diagrams

Data Flow Diagram (DFD)

The Data Flow Diagram shows how information moves between the users, the system, and the database.

- Teacher → System → Database: Enters attendance, grades, updates student info, views reports.

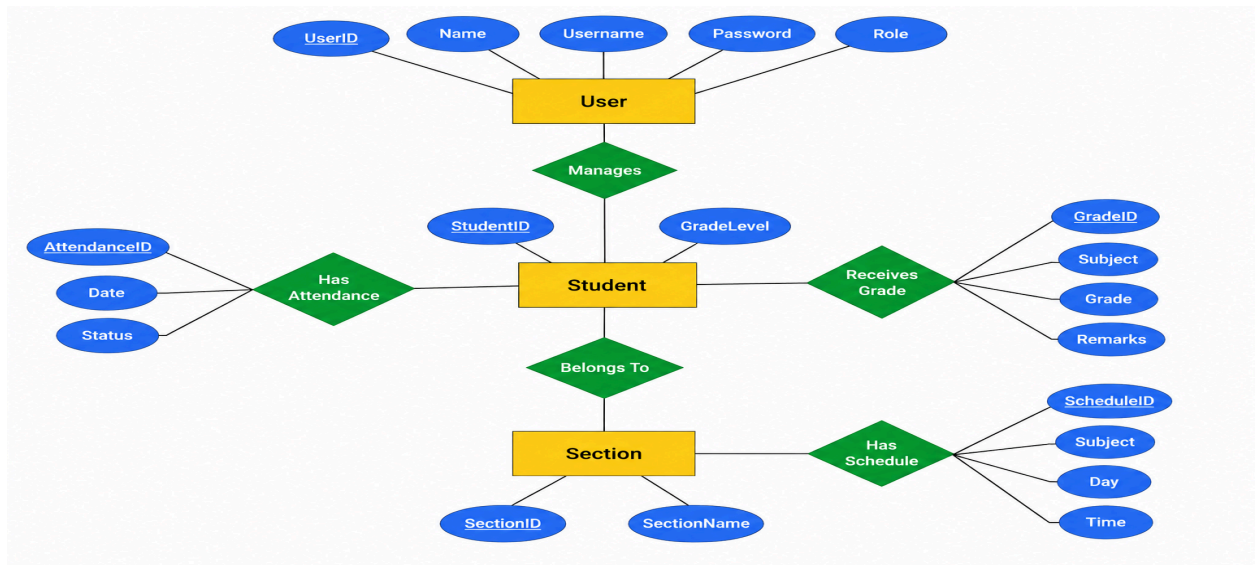
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- Administrator → System → Database: Manages users, student records, sections, schedules, generates reports.
- System: Validates, processes, organizes data.
- MySQL Database: Stores all records securely.
- Outputs: Attendance reports, grade reports, behavioral reports, student records, schedule & section information.

Entity Relationship Diagram (ERD)

Shows database entities and relationships:



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USE CASE DIAGRAM

Actors

- Administrator
- Teacher

Administrator Use Cases

- Log In
- Manage User Accounts
- Manage Student Records
- Manage Sections
- Manage Schedules
- Generate Reports
- View Behavioral Analysis

Teacher Use Cases

- Log In

- Record Attendance
- Input Grades
- View Student Records
- View Student Schedules
- Generate Reports
- View Behavioral Analysis

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

Overview of the Developed System

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The Web-Based Student Behavioral Pattern System is a web-based application developed to automate and improve the management of student information, attendance monitoring, grade tracking, and behavioral analysis for Santa Clarita Integrated School. The system integrates student profile management, daily attendance recording, academic grade monitoring, class scheduling, section assignments, and behavioral pattern analysis into a single, secure, and browser-accessible platform. It enables school administrators and teachers to efficiently maintain student records, track daily attendance, encode academic grades, manage class assignments, and generate reports that support student monitoring and evidence-based decision-making.

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Unlike the school's previous manual process of recording student information, attendance, grades, and schedules using paper logbooks, scattered spreadsheets, and semi-computerized files, the developed system centralizes all student-related information in a secure, structured database. The application minimizes human errors such as duplicate entries, missing records, and inconsistent updates; eliminates redundant record-keeping tasks; and provides real-time access to organized student data. It also generates summaries, trend analyses, and behavioral alerts that help school personnel identify students who may need academic guidance or behavioral intervention—allowing for timely support rather than delayed responses.

The system was developed to provide Santa Clarita Integrated School with an affordable, user-friendly, and sustainable digital solution that improves record management accuracy, reduces administrative workload, and strengthens student monitoring capabilities. By correlating attendance patterns with academic performance, the platform transforms daily collected data into actionable insights, moving the school from simple record-keeping to meaningful behavioral monitoring.

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The major objectives of the developed system are as follows:

- Centralize student records in one secure and organized database.
- Provide accurate daily attendance recording and monitoring with easy retrieval.
- Enable systematic recording, updating, and tracking of student academic grades.
- Integrate class schedules and section assignments for organized student grouping.
- Identify behavioral patterns by correlating attendance trends with academic performance.
- Reduce manual record-keeping time and improve information retrieval efficiency.

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- Generate clear reports and summaries to support timely intervention and decision-making.
- Ensure the system is accessible, reliable, and suitable for daily use by school personnel.

The development followed modern software engineering principles: responsive user interface design, secure role-based authentication, centralized database management, data validation, and reliable cross-module data processing. These features ensure that authorized users—Administrators, Teachers, and Students—access only the functions appropriate to their roles while maintaining the accuracy, integrity, and confidentiality of all student information.

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The completed system consists of several integrated modules that work together to capture, process, store, and analyze student-related data. These modules include User Authentication, Student Record Management, Attendance Monitoring, Grade Management, Schedule and Section Management, Behavioral Pattern Analysis, Report Generation, and Dashboard Analytics. Together, these components provide a comprehensive platform that streamlines daily school operations and strengthens the school's ability to monitor and support learner progress.

System Implementation

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This phase involved the full development and configuration of the Web-Based Student Behavioral Pattern System. The system was built using HTML5, CSS3, JavaScript, PHP, and MySQL, following the file structure and modular organization found in the repository. It includes configuration files for application settings and database connection, shared components for consistent page layout and security, reusable database operation functions, and role-specific modules for administrators, teachers, and students. The implemented system centralizes data storage, enforces secure role-based access, and connects attendance and grade records to support behavioral analysis.

Configuration and Core Components

The system is structured around two core configuration files that establish the environment:

- `app.php` – Defines application-wide constants such as the site name, base URL, file upload paths, maximum file size, allowed image formats, session duration, pagination limits, and standard date formats. It also automatically creates required storage directories if they do not exist.

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- `database.php` – Establishes the database connection using PDO with prepared statements for security. It defines connection parameters (host, database name, username, password, character set) and implements a singleton connection class to ensure efficient, consistent database access across all modules.

Shared includes provide reusable functionality across the entire system:

- `auth.php` – Manages user sessions, login/logout, role verification, session timeout, and access restriction. It ensures pages are only accessible to logged-in users and enforces permission levels (Administrator / Teacher / Student).

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- `db_operations.php` – Contains over 90 reusable database functions covering student records, course management, sections, enrollments, attendance tracking, grade computation, assessments, assignments, announcements, messaging, registration approvals, and dashboard statistics. All database queries use prepared statements to prevent SQL injection.
- `helpers.php` – Provides utility functions for formatting dates, sanitizing input, generating page titles, and other shared tasks.
- `header.php` / `footer.php` – Standardized page components that ensure consistent layout, navigation, and Bootstrap styling across all pages.

Admin Module

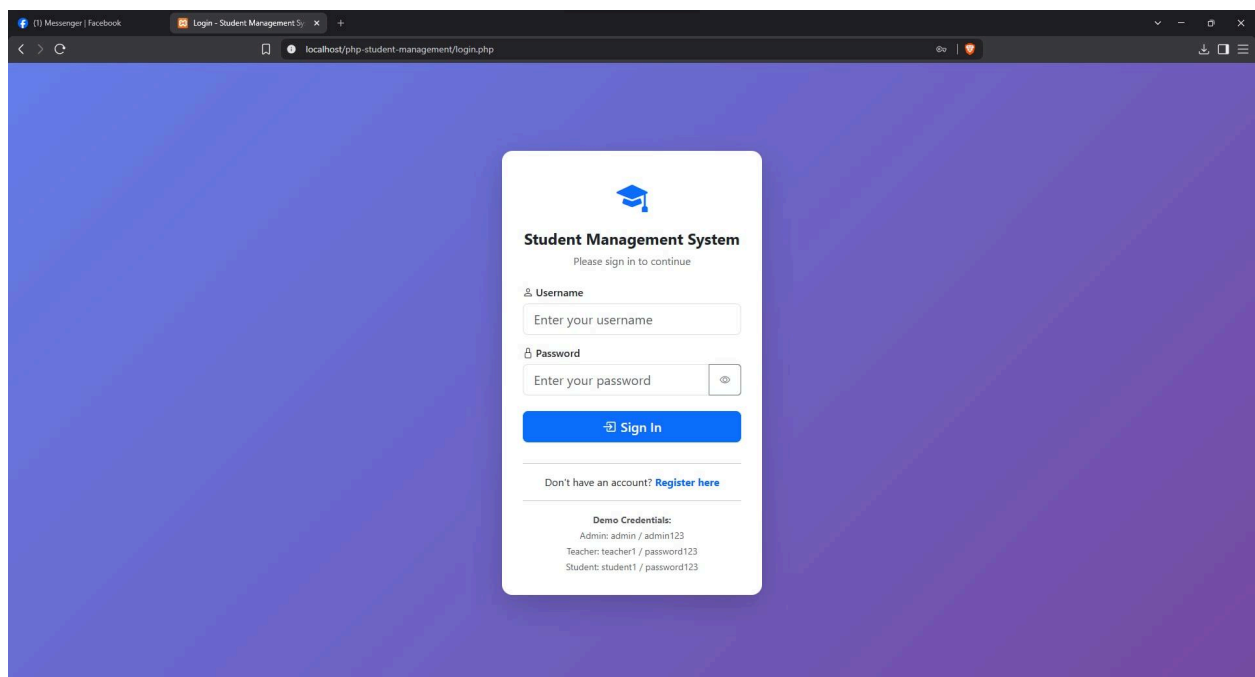
Login Module

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The Login Module serves as the primary access point for all system users. It authenticates users by verifying their credentials against securely hashed passwords, then redirects them to the appropriate dashboard based on their assigned role. The module enforces session security, automatically logs users out after one hour of inactivity, and protects all internal pages from unauthorized access.

Figure 4.1 – Login Interface of the Web-Based Student Behavioral Pattern System



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As shown in Figure 4.1, the login interface provides a clean, straightforward form where users enter their credentials. Upon successful authentication, the system stores the user's ID, full name, role, and login time in an encrypted session. Users are then directed to their role-specific dashboard. Invalid attempts show a clear error message without revealing which field was incorrect.

Dashboard Module

The Dashboard Module is the main landing page displayed after successful login. It presents a role-specific overview with summary statistics, quick-access navigation, and notifications.

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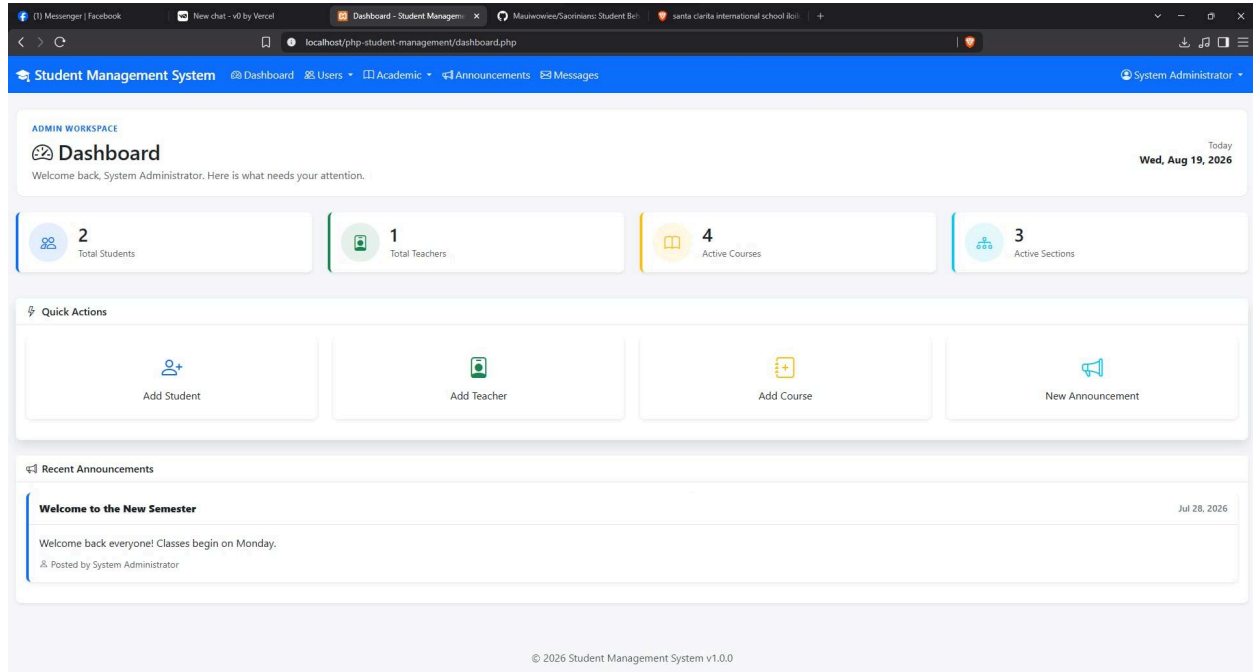
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- Administrator Dashboard – Displays total counts for students, teachers, active courses, active sections, and current enrollments. It provides direct links to manage users, courses, sections, enrollment requests, pending registrations, announcements, and system settings.
- Teacher Dashboard – Shows the teacher's assigned sections, number of enrolled students per class, and quick actions for attendance, grades, and announcements.
- Student Dashboard – Displays enrolled courses, current schedule, recent grades, attendance summary, and system announcements.

Figure 4.2 – Administrator Dashboard Interface

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As shown in Figure 4.2, the dashboard is organized with a consistent header, sidebar navigation, and main content area. Summary cards provide at-a-glance information, while clearly labeled menu items guide users to each module. This centralized design allows personnel to quickly assess key indicators and navigate to needed functions without searching through multiple pages.

Student Record Management Module

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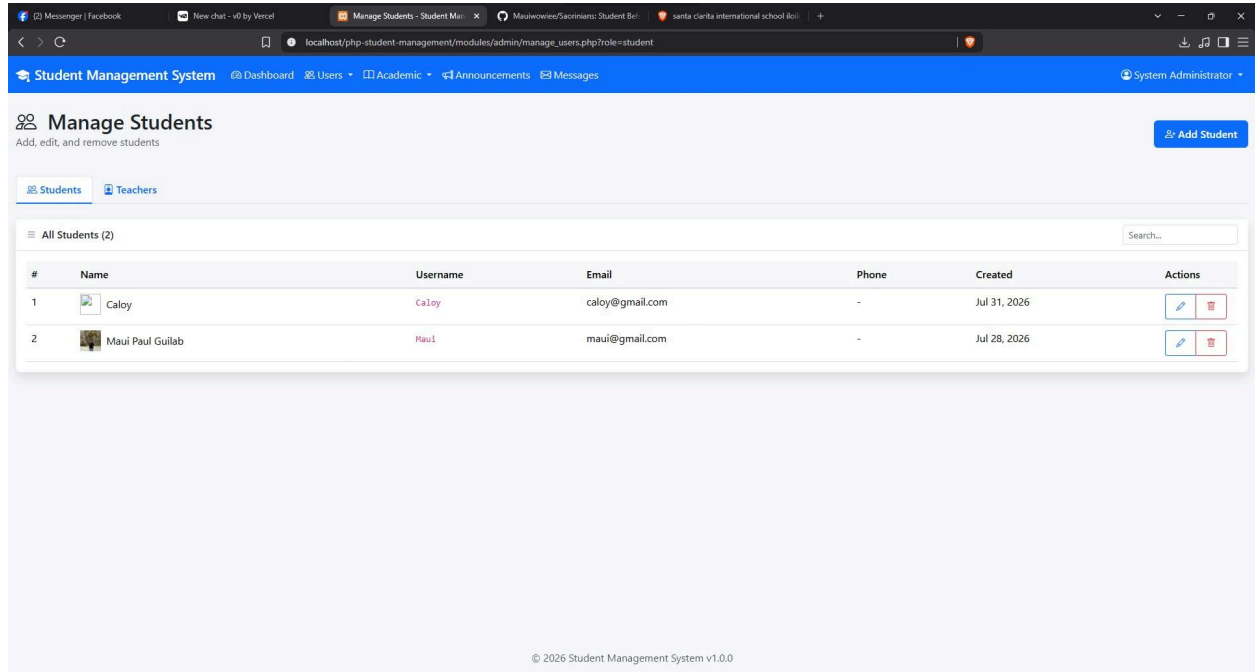
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This module allows the Administrator to create, view, edit, and deactivate student accounts and profiles. Each student record stores unique identification, full name, username, hashed password, assigned role, email address, contact information, home address, date of birth, and optional profile picture. Profiles are automatically created when an account is added and can be updated at any time.

Figure 4.3 – Student Record Management Module

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As shown in Figure 4.3, the interface displays all students in a searchable, sortable table. Users can filter by name or ID, add new students individually or through approved registrations, and update information directly. The system validates entries to prevent duplicates and ensures required fields are complete before saving.

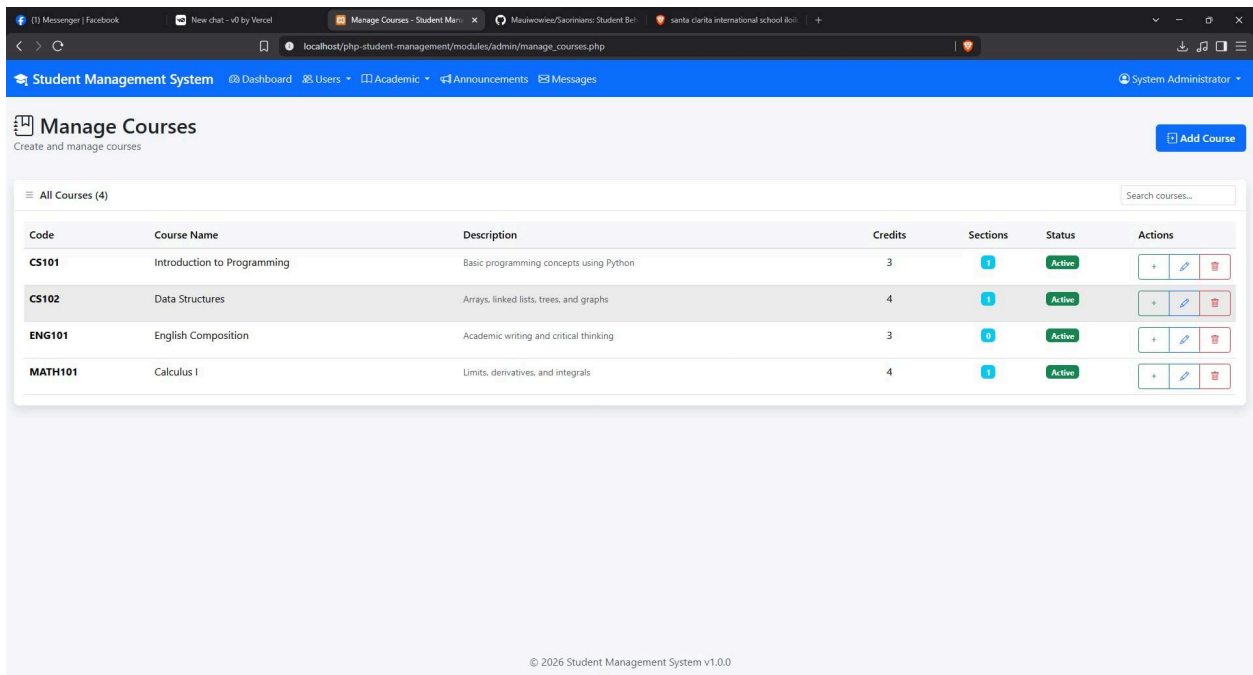
Course and Section Management Module

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This module manages academic courses and class sections. Administrators create course records with codes, names, descriptions, and credit units. Sections are then created under each course, assigning the teacher, schedule, room number, and maximum student capacity.

Figure 4.4 – Course and Section Management Module



Code	Course Name	Description	Credits	Sections	Status	Actions
CS101	Introduction to Programming	Basic programming concepts using Python	3	1	Active	+ ✎ ✖
CS102	Data Structures	Arrays, linked lists, trees, and graphs	4	1	Active	+ ✎ ✖
ENG101	English Composition	Academic writing and critical thinking	3	0	Active	+ ✎ ✖
MATH101	Calculus I	Limits, derivatives, and integrals	4	1	Active	+ ✎ ✖

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As shown in Figure 4.4, courses are listed alphabetically, and clicking a course displays its sections. Each section shows the assigned teacher, schedule, room, and current enrollment count. This structure ensures that every class grouping is clearly defined and linked to the correct course and instructor.

Attendance Monitoring Module

The Attendance Module allows Teachers to record daily attendance per section. Teachers can mark all students Present or Absent with one click, then adjust individual entries as needed. The system stores each record with date, status, remarks, and the person who marked it.

Figure 4.5 – Attendance Monitoring Interface

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Take Attendance
Mark attendance for your sections

Select Section & Date

Section: CS101 - Section-A

Date: 19/08/2026

[Load Students](#)

Section Info
Course: Introduction to Programming
Room: Room 101
Schedule: MWF 9:00-10:00 AM

Attendance for CS101 - Section-A (Aug 19, 2026)

[All Present](#) [All Absent](#)

#	Student	Status	Remarks
1	Caloy @Caloy	Present	Optional remarks
2	Maui Paul Gullab @Maui	Present	Optional remarks

[Cancel](#) [Save Attendance](#)

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As shown in Figure 4.5, the interface displays all enrolled students for the selected section on the chosen date. Attendance status options include Present, Absent, Late, or Excused. Saved records are immediately available in the student's attendance history and used in behavioral calculations.

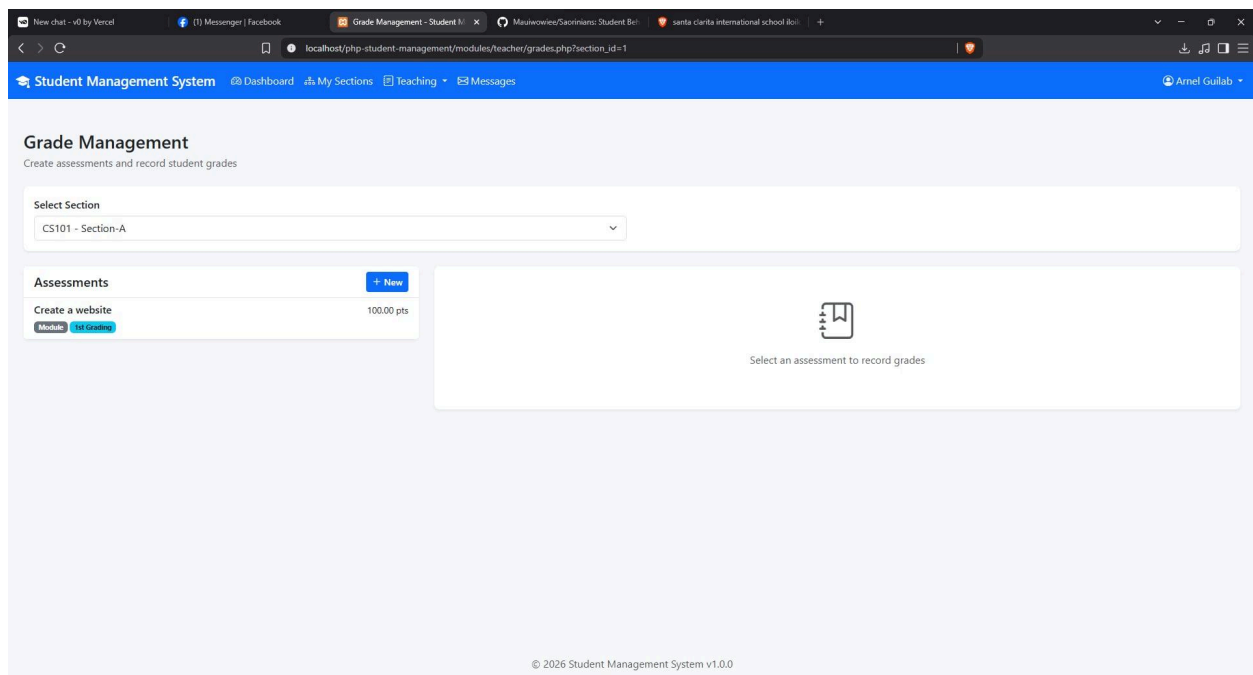
Grade and Assessment Module

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This module supports recording academic performance through assessments (quizzes, tests, exams, projects) and assignments. Teachers define each assessment's maximum score, weight, and due date, then record individual student scores. The system automatically computes weighted averages and period grades.

Figure 4.6 – Grade and Assessment Management Interface



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As shown in Figure 4.6, scores are entered per assessment or per student, with the system calculating component averages and overall grades. Grading periods follow the standard four-quarter system, and computed grades are visible to authorized personnel with proper role-based permissions.

Behavioral Pattern Analysis Module

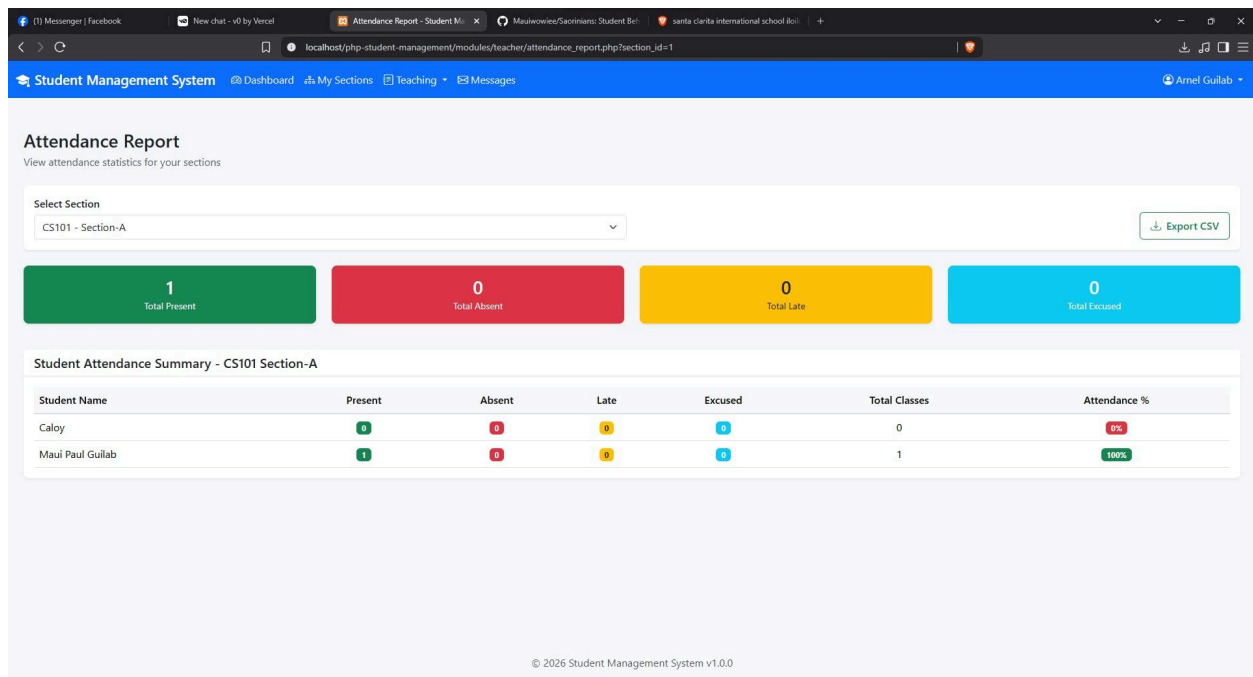
This is the core analytical feature of the system. It cross-references attendance history with academic performance to identify trends and flag students who may be at risk. The system calculates:

- Attendance percentage – Present days divided by total school days
- Attendance trend – Improving, stable, or declining
- Grade average and trend – Current performance and direction of change
- Combined risk indicator – Flags students with falling attendance AND declining grades

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Figure 4.7 – Behavioral Pattern Analysis Report



As shown in Figure 4.7, the report lists students with their attendance percentage, current grade average, attendance trend, grade trend, and a clear risk indicator. Students showing both declining attendance and falling grades are highlighted for early intervention. This transforms separate attendance and grade records into actionable monitoring information.

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Report Generation Module

The Reports Module exports summaries and records for printing or sharing. Available reports include:

- Student profile and enrollment history
- Attendance summary per section or per student
- Grade summary and progress report
- Behavioral pattern and risk alert report
- Class roster and schedule list

Figure 4.8 – Report Generation Interface

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The screenshot shows a web browser window displaying the 'Student Management System' interface. The page title is 'Enrollment Management' with a subtitle 'Enroll students in course sections'. The interface is divided into two main sections: 'Select Section' and 'Enrolled Students'.

Select Section:

- Course Section:** A dropdown menu showing 'CS101 - Section-A (Arnel Guilab)'.
- Section Details:**
 - Course:** CS101 - Introduction to Programming
 - Section:** Section-A
 - Teacher:** Arnel Guilab
 - Room:** Room 101
 - Schedule:** MWF 9:00-10:00 AM
 - Enrolled:** 2 / 40

Enrolled Students:

#	Student	Email	Enrolled Date	Grade	Actions
1	 Caloy @Caloy	caloy@gmail.com	Jul 31, 2026	-	
2	 Maui Paul Guilab @Maui	maui@gmail.com	Jul 28, 2026	-	

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As shown in Figure 4.8, users select the report type, section or student, and date range. The system generates a formatted, printable view that can also be exported or copied for official use. Reports are automatically organized and labeled for documentation purposes

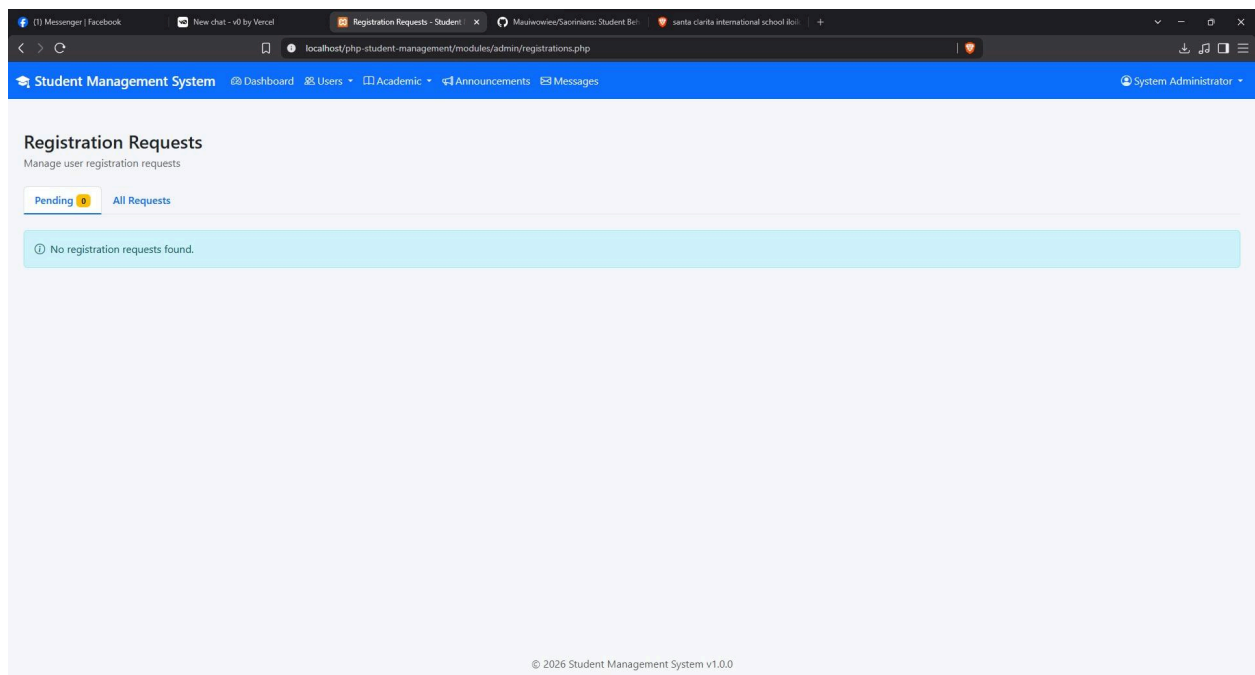
User and Registration Management Module

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Administrators manage all system accounts and review pending registration requests. Prospective users may submit an online request; the Administrator reviews, approves or rejects, and the system automatically creates the account and profile upon approval.

Figure 4.9 – Registration and User Management Interface



As shown in Figure 4.9, pending requests display applicant details and credentials. Approved requests become active system accounts, while rejected requests are logged with remarks. This module ensures only authorized persons gain system access while maintaining a clear audit trail.

Testing Results

Testing was conducted throughout development to verify that every function operated correctly and that modules worked together as intended. The process followed four standard testing stages: Unit Testing, Integration Testing, System Testing, and User Acceptance Testing. Issues discovered were corrected and retested before proceeding.

Summary of Testing Activities

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Testing Type	Performed By	Focus of Testing	Result / Observation
Unit Testing	Development Team	Individual functions and modules	Functions tested separately; issues fixed and retested.
Integration Testing	Development Team	Data flow between connected modules	Information passed correctly between modules after fixes.
System Testing	Development Team	Complete end-to-end system operation	Full workflow verified; major functions all operational.

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User Acceptance Testing	School Administrators & Teachers	Real-world usability and requirements alignment	Feedback incorporated; system refined to match actual needs.
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Unit Testing Results

Unit testing examined each function and module independently to ensure it performed exactly as designed.

Module / Function	Test Performed	Expected Result	Observed Result	Status
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User Authentication	Log in with valid credentials	User authenticated and directed to dashboard	Login successful; session created correctly	Passed
Student Records	Add, edit, or delete a student	Record saved, updated, or removed correctly	Database updated; changes reflected immediately	Passed
Course & Section Management	Create and assign sections to courses	Section linked correctly to course and teacher	Relationships maintained; sections appear under correct course	Passed

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Attendance Recording	Mark attendance and save	Records stored and available in history	Attendance saved; totals and percentages computed	Passed
Grade & Assessment	Record scores and compute grade	Weighted average calculated correctly	Grades computed; weights applied per rules	Passed
Behavioral Analysis	View attendance vs grade trends	System generates attendance %, trends, and alerts	Analysis generated; at-risk students flagged correctly	Passed

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Report Generation	Generate and preview report	Formatted report displays correct data	Report matches source records; layout print-ready	Passed
Access Control	Attempt restricted page without permission	Access denied; redirected to login	Unauthorized access blocked; role permissions enforced	Passed

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Integration Testing Results

Integration testing verified that modules shared data correctly and that changes in one module updated related information elsewhere.

Integrated Functions	Test Performed	Expected Result	Observed Result	Status
Enrollment ↔ Attendance & Grades	Enroll student → appears in attendance and grade sheets	Student appears in all related modules automatically	Appears correctly across modules	Passed

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Attendance ↔ Behavioral Analysis	Record attendance → update attendance % and trends	Analysis refreshes to reflect new attendance record	Trends update immediately	Passed
Grades ↔ Behavioral Analysis	Record grade → update grade average and trend	Analysis refreshes to reflect new grade	Grade trend updates correctly	Passed
Sections ↔ Reports	Create section → appears in report options	Section available in all relevant reports	Section listed correctly	Passed

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Role-Based Access	Log in as different roles → see correct menu	Each role sees only permitted functions	Menus match assigned permissions	Passed
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System Testing Results

System testing evaluated the complete working application to confirm the full workflow operated reliably.

Feature Area	Test Performed	Expected Result	Observed Result	Status
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Full Workflow	Create course → create section → enroll student → mark attendance → enter grades → view analysis → generate report	Complete sequence functions without error	Full sequence completed successfull y	Passed
Data Consisten cy	Update student name → check all related records	Name updates everywhere consistentl y	Consistent across all modules	Passed
Security	View database files directly; bypass login	Access denied or restricted	Direct access blocked; login required	Passed

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Reliability	System used continuously for several days	Functions remain available; no data loss	Stable operation; sessions expire correctly	Passed
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User Acceptance Testing Results

School administrators and teachers reviewed the system and provided feedback during development. Their suggestions were incorporated into the final version.

Area Reviewed	Expected Result	Observed Result	Status
Student Record Management	Easy to add, find, and update student information	Interface clear; search works well; fields complete	Accepted

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Attendance Monitoring	Fast daily recording; easy to review history	Marking attendance takes under 2 minutes per section	Accepted
Grade Management	Encoding and computing grades matches school practice	Grading periods and computation align with school system	Accepted
Behavioral Analysis	Clear indicators help identify students needing attention	Flags and trends easy to understand; helpful for guidance	Accepted
Report Generation	Reports ready for printing and filing	Format suitable for official use	Accepted

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Ease of Use	Interface understandable without extensive training	Teachers and staff learned basic functions quickly	Accepted
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Hardware and Software Requirements

Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 or equivalent	Intel Core i5 or higher
Memory (RAM)	4 GB	8 GB or higher
Storage	256 GB HDD	256 GB SSD or higher

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Display	1366 × 768 resolution	Full HD 1920 × 1080 display
Network	Stable internet or local network	Local area network with internet backup

Software Requirements

Software / Technology	Purpose
Windows 10 / 11 or compatible OS	Server or workstation operating system
PHP 7.4+	Backend scripting and logic
MySQL 5.7+	Relational database management

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XAMPP / WAMP	Local web server and database environment
HTML5, CSS3, JavaScript	Frontend structure, styling, and interactivity
Bootstrap 5	Responsive interface components
Visual Studio Code	Code editing
Google Chrome / Microsoft Edge	Web browser for system access

CHAPTER V

CONCLUSION AND RECOMMENDATIONS

INTRODUCTION

This chapter presents the summary of findings, conclusions, and recommendations of the study entitled Web-Based Student Behavioral Pattern System for Santa Clarita Integrated School. It summarizes the key results obtained during the development, implementation, and evaluation of the proposed system. The chapter also determines whether the objectives of the study were successfully achieved based on the findings gathered from testing and user evaluation. Finally, it provides recommendations for future improvements and enhancements that can further increase the system's efficiency, functionality, and overall effectiveness in supporting the daily administrative and instructional operations of Santa Clarita Integrated School.

SUMMARY OF FINDINGS

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The development and evaluation of the Web-Based Student Behavioral Pattern System yielded significant results that aligned directly with the general and specific objectives of the study. Based on the results of unit testing, integration testing, system testing, and user acceptance conducted by the researchers and the selected school personnel, the following major findings were observed:

1. Centralized Student Record Management – The system successfully consolidated student profiles, personal information, enrollment details, and academic records into a single, secure, and searchable database. Administrators and teachers were able to retrieve complete student information in seconds, compared to the manual process which previously took several minutes or even hours to locate scattered records. Duplicate entries, inconsistent updates, and missing documents were significantly reduced because all authorized users accessed the same centralized source of information.

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2. Attendance Monitoring and Tracking – The Attendance Monitoring Module enabled teachers to record daily attendance per section in less than two minutes, compared to the 10-15 minutes required using manual logbooks. The system automatically computed attendance percentages, generated daily and monthly summaries, and maintained a complete attendance history per student. It also provided clear indicators of attendance trends, making it easier to identify students with chronic absenteeism or recurring tardiness before these issues affected academic performance.
3. Academic Grade and Performance Tracking – The Grade and Assessment Module allowed teachers to record scores for quizzes, tests, exams, projects, and assignments. The system automatically computed weighted averages and period grades based on defined grading components and weights. Grades were stored securely and linked directly to student records, enabling authorized users to view academic progress over time and detect declining performance early.

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4. Behavioral Pattern Identification and Analysis – The core feature of the system – Behavioral Pattern Analysis – successfully correlated attendance records with academic grades to generate meaningful insights. By combining attendance percentages, attendance trends, grade averages, and grade trends, the system was able to identify students showing signs of disengagement or academic difficulty. Students with both declining attendance and falling grades were automatically flagged, allowing teachers and administrators to provide timely guidance and intervention. This feature demonstrated that attendance and academic performance, when analyzed together, serve as reliable early indicators of student welfare and engagement.

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5. Class Schedule and Section Management – The system

integrated course management, section assignments, teacher allocation, room assignments, and schedule information into one platform. Administrators could easily create and update course offerings, assign teachers and schedules to sections, and view class rosters. Enrolled students were automatically linked to their assigned sections, making attendance recording and grade encoding faster and more organized.

6. Report Generation and Decision Support – The Report

Generation Module produced organized, formatted, and printable reports including student profiles, attendance summaries, grade progress reports, class rosters, schedule matrices, and behavioral pattern alerts. Reports were generated in seconds rather than the hours previously required for manual preparation, enabling school personnel to submit required documentation on time and make evidence-based decisions.

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7. System Usability, Security, and Reliability – Evaluation

results from school administrators and teachers indicated that the system is user-friendly, secure, and reliable.

Role-based access control ensured that only authorized persons could access specific pages and functions –

administrators managed system settings and all records;

teachers handled attendance, grades, and their assigned sections; and students viewed only their own records.

Passwords were securely hashed, sessions expired

automatically after inactivity, and all database queries

used prepared statements to prevent unauthorized access and data loss. The system operated consistently across

different web browsers and devices, with data remaining

accurate and intact throughout the evaluation period.

CONCLUSION

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The Web-Based Student Behavioral Pattern System proved to be a practical, functional, and effective solution to the operational challenges faced by Santa Clarita Integrated School. By replacing manual logbooks, scattered spreadsheets, and fragmented record-keeping processes with a centralized web-based platform, the system successfully reduced administrative workload, improved data accuracy, and strengthened the school's ability to monitor student progress and behavior.

The study achieved all stated objectives: it centralized student records, streamlined attendance monitoring, systematized grade tracking, organized schedules and sections, enabled behavioral pattern analysis, improved information retrieval efficiency, generated useful reports, and provided a secure, user-friendly environment suitable for daily school use. The development followed standard software engineering practices – including modular design, secure authentication, database normalization, and iterative testing – resulting in a system that is stable, scalable, and maintainable.

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Most importantly, the study demonstrated that simple but meaningful analysis of data already collected by the school – specifically attendance and academic performance – can produce powerful insights that support early intervention. The system did not merely store records; it transformed daily collected information into actionable knowledge that helps teachers and administrators identify students in need of support before problems become serious. This moves the school from reactive record-keeping to proactive student monitoring, directly addressing the core problem identified at the beginning of the study.

Based on the evaluation results and the findings from all testing stages, the research questions were answered positively. The proposed system successfully centralized student information, improved attendance and grade monitoring, integrated schedules and sections, identified behavioral patterns through data correlation, streamlined daily operations, and generated reports that support decision-making. The system is therefore confirmed to be a suitable, sustainable, and cost-effective digital solution aligned with the school's needs and the Department of Education's DigiEd 2028 roadmap.

RECOMMENDATIONS

To further enhance the functionality, capacity, and long-term value of the Web-Based Student Behavioral Pattern System, the researchers propose the following recommendations, organized into System Improvements, Future Research Directions, and Implementation Strategies:

System Improvements

1. Enhanced Behavioral Prediction – Expand the behavioral analysis module to include longer trend analysis, historical comparison across grading periods, and risk-level scoring that helps prioritize students requiring immediate attention. This would provide even clearer guidance to counselors and administrators.

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2. Parent and Guardian Access Portal – Develop a restricted-access portal so parents and guardians can view their child's attendance summary, grades, schedule, and announcements without accessing other students' information. This strengthens home-school communication and keeps families informed of student progress.
3. Automated Notification System – Integrate email or SMS notifications to alert teachers, guidance counselors, or parents when a student reaches a defined threshold of absences or shows a significant drop in academic performance. Timely alerts can greatly improve intervention outcomes.
4. Advanced Report Export and Formatting – Enable reports to be downloaded or exported in standard formats such as PDF, Excel, and CSV. This would make it easier for the school to share, archive, and further process information in the formats required by DepEd or other stakeholders.

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5. Data Visualization and Dashboard Enhancements – Add charts, graphs, and trend lines to the dashboard so that attendance patterns, grade distributions, and student progress can be understood visually at a glance. Visual summaries help make data more accessible and easier to explain during meetings.
6. Mobile-Friendly Interface and Application – Further optimize the system interface for mobile devices or develop a dedicated mobile version. This would allow teachers and administrators to record attendance, check records, and view reports conveniently using smartphones or tablets.
7. Two-Factor Authentication and Enhanced Security – Add optional two-factor authentication for administrator accounts and implement automatic periodic password updates. This further strengthens data protection and ensures full compliance with the Data Privacy Act of 2012.
8. Assignment and Submission Management – Expand the assessment module to allow teachers to post assignments, accept online submissions, and grade work directly within the system. This connects academic tasks directly to student performance records.

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Future Research Directions

1. Integration with Learning Management Systems – Future researchers may explore connecting the system to an online learning platform, so that class activities, resources, and student engagement can be tracked alongside attendance and grades for a more complete picture of student behavior.
2. Predictive Analytics Using Machine Learning – Advanced studies may apply machine learning algorithms to historical data to predict which students are most likely to struggle academically or behaviorally. This can improve early warning accuracy beyond what simple trend analysis can provide.
3. Long-Term Impact Evaluation – Conduct a longitudinal study to measure how the system affects student attendance, academic performance, and intervention outcomes over one or more full school years. This would provide deeper understanding of how behavioral monitoring influences student success.

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4. Adaptation for Wider School Use – Develop and evaluate the system for use across multiple grade levels, larger student populations, or different school types. This can help identify whether the same design works effectively in varied educational settings.
5. Cost-Benefit Analysis of Digital Systems – Conduct a formal study comparing the time saved, errors reduced, and resources freed by using the system versus manual methods. This can help other schools justify adopting similar digital solutions.

Implementation Strategies

1. Comprehensive User Training – Before full deployment, conduct orientation sessions and provide printed quick-reference guides for all administrators and teachers. Hands-on training ensures that users understand how to operate each module properly and helps them transition smoothly from manual methods to the new system.

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2. Regular Data Backup and Maintenance – Establish a routine schedule for database backups – ideally daily – and store backup copies securely. Perform periodic system maintenance, including updates, security patches, and error checks, to ensure continuous reliable operation.
3. Continuous Feedback and Improvement – Maintain an open channel for user suggestions and feature requests. The Agile methodology used in development allows the system to grow and improve based on actual daily use. Regular updates should address issues reported and introduce features that further fit the school's evolving needs.
4. Clear Data Privacy and Access Policies – Establish written policies defining who may access what information, how data is stored and protected, and how long records are retained. This ensures compliance with the Data Privacy Act of 2012 and builds trust among students, parents, and staff.

5. Phased Deployment – Roll out the system by stages: begin with attendance and student records, then add grades and scheduling, and finally activate behavioral analysis and reports. Phased implementation allows users to become comfortable with each part before moving to the next, reducing confusion and building confidence.

IMPACT OF THE STUDY

This study contributes to the growing body of knowledge regarding digital transformation in Philippine education. It demonstrates that a well-designed, affordable web-based information system — built using standard tools such as HTML5, CSS3, JavaScript, PHP, and MySQL — can effectively replace inefficient manual processes and significantly improve how a school manages student information. The system is not only functional but also practical, sustainable, and tailored to the specific workflows and needs of Santa Clarita Integrated School.

More importantly, the study highlights the value of using data already collected by schools to improve student support. By connecting attendance records with academic performance into a single coherent view, the system transforms separate pieces of information into a powerful tool for understanding student behavior. This supports the idea that schools do not always need complex or expensive technology to make a meaningful difference — they simply need the right way to organize, connect, and use the information they already have.

The project also serves as a useful reference for other educational institutions considering digital transformation. It shows that centralized, web-based systems improve record accuracy, reduce workload, and strengthen decision-making — while remaining accessible, secure, and compliant with national standards such as the Data Privacy Act of 2012 and the DigiEd 2028 roadmap.

SUMMARY

In summary, the Web-Based Student Behavioral Pattern System represents the successful application of software development principles, educational research, and iterative design to solve real operational challenges at Santa Clarita Integrated School. By centralizing student records, automating attendance monitoring, systematizing grade management, organizing schedules and sections, analyzing behavioral patterns, and generating useful reports, the system effectively addresses every problem identified in the study. It reduces manual labor, minimizes recording errors, and most importantly, helps school personnel identify and support students who may be struggling — early enough to make a real difference.

While the system has limitations — such as requiring reliable internet access, regular data maintenance, and ongoing user training — the evaluation results confirm that it meets or exceeds all requirements set at the beginning of the study. It is functional, usable, reliable, secure, and efficient. As a whole, this project demonstrates that custom-developed systems built specifically for an institution's actual needs are not only feasible but also highly effective tools for improving educational management and student support.

This study serves as a valuable contribution to the field of educational technology and IT capstone project research. It provides a working example, documented design, and tested implementation that future researchers, developers, and schools may adapt and improve upon. Ultimately, it stands as proof that technology, when designed properly and grounded in real needs, can become a powerful partner in helping schools serve their learners better.

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